

Bayesian Inference in systems biology

Theory

What is Bayesian inference?

Probability 101

Bayes' theorem in biology

Priors and likelihood

Interpretation of posteriors

Markov Chain Monte Carlo

Applications

Population dynamics in
microscopic and
macroscopic communities

Phylogeny and model
selection

ODE generative models:
Morphogen patterning of
embryonic tissues

Bayes' Theorem

$$P(X, Y) = P(X)P(Y|X)$$

$$P(X, Y) = P(Y)P(X|Y)$$

$$\frac{\cancel{P(X, Y)}}{\cancel{P(X, Y)}} = \frac{P(X)P(Y|X)}{P(Y)P(X|Y)}$$

$$P(X|Y) = \frac{P(X)P(Y|X)}{P(Y)}$$

Bayesian Inference

$$P(X|Y) = \frac{P(X)P(Y|X)}{P(Y)}$$



Prior probability

Likelihood

$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

Posterior probability

Y : Experimental data

X : Parameter value (δ)

Bayesian Inference

$$P(X|Y) = \frac{P(X)P(Y|X)}{P(Y)}$$

Likelihood

Prior
probability

$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

Posterior
probability

What about $P(data)$? ✓

How do I choose the prior? ✓

What am I supposed to do with the posterior? ✓

Okay, but how do apply this with my laptop? ✓

Can you give examples of the likelihood? ✓

What about $P(data)$? ✓

$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

→ Marginal Likelihood ←

$$P(data) = \int P(data, \delta) d\delta = \int P(data|\delta)P(\delta) d\delta$$

Let's compare the posterior probabilities of two values :

$$\frac{P(\delta = \delta_1 | data)}{P(\delta = \delta_2 | data)} = \frac{\frac{P(\delta_1)P(data|\delta_1)}{\cancel{P(data)}}}{\frac{P(\delta_2)P(data|\delta_2)}{\cancel{P(data)}}}$$

What about $P(data)$? ✓✓

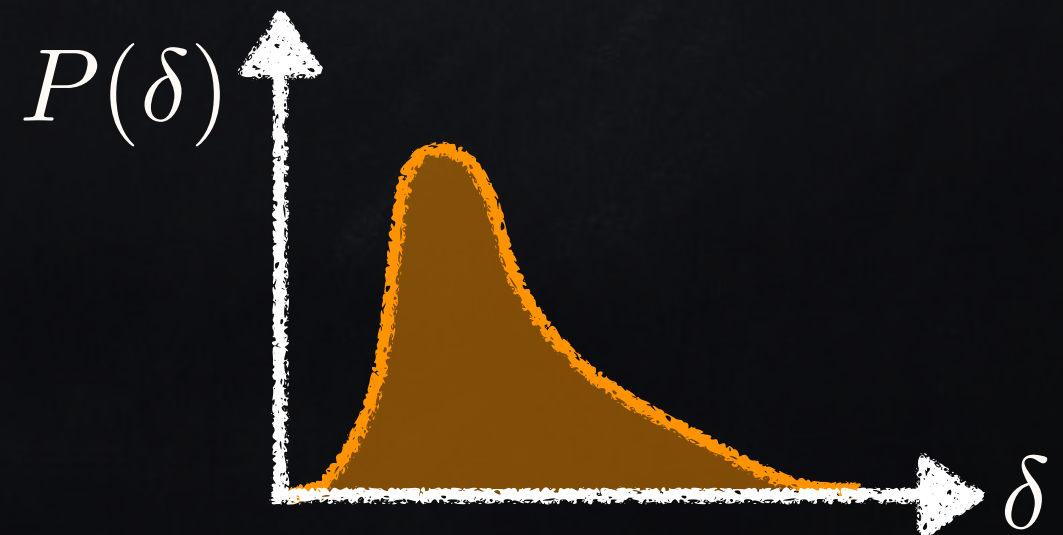
$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

In practice is not necessary 🍹

How do I choose the prior? ✓

Reflecting your prior knowledge 🧐

- Noninformative priors
- Previous studies
- Previous Bayesian analysis



What about $P(data)$? ✓✓

$$P(\delta)P(data|\delta)$$

$ta)$

How do I choose the prior

Refle

Question:

- Two different studies report that the mean diameter of the trunk of oaks in a particular forest

$$D = 0.7 \pm 0.4 \text{ m}$$

$$D = 1.0 \pm 0.1 \text{ m}$$

How would you encode this in a prior distribution?
What if the studies report:

$$D = 0.7 \pm 0.1 \text{ m}$$

$$D = 1.0 \pm 0.1 \text{ m}$$

- Noninformative priors
- Previous studies
- Previous Bayesian analysis

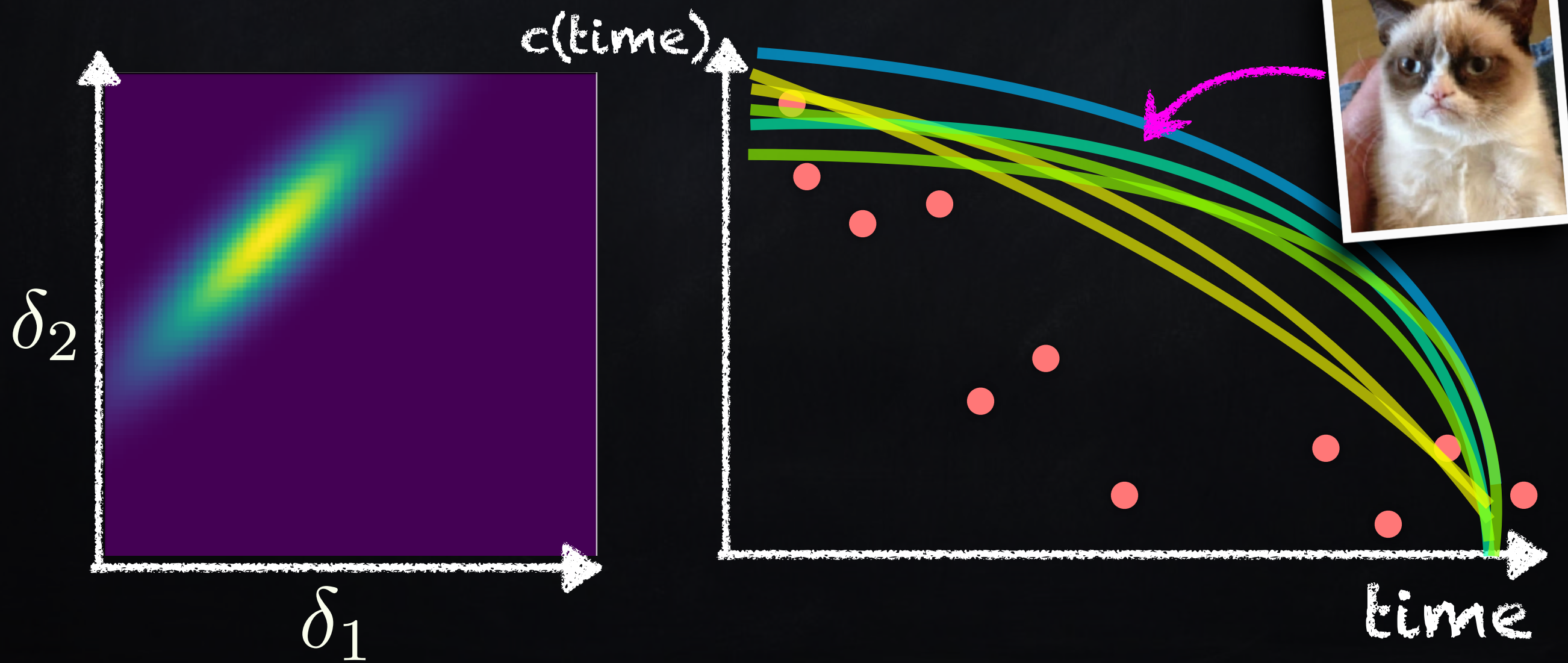
What am I supposed to do with the posterior? ✓

First, we need to check that the results make sense

Posterior predictive checks

Correct priors?

Correct likelihood/model?



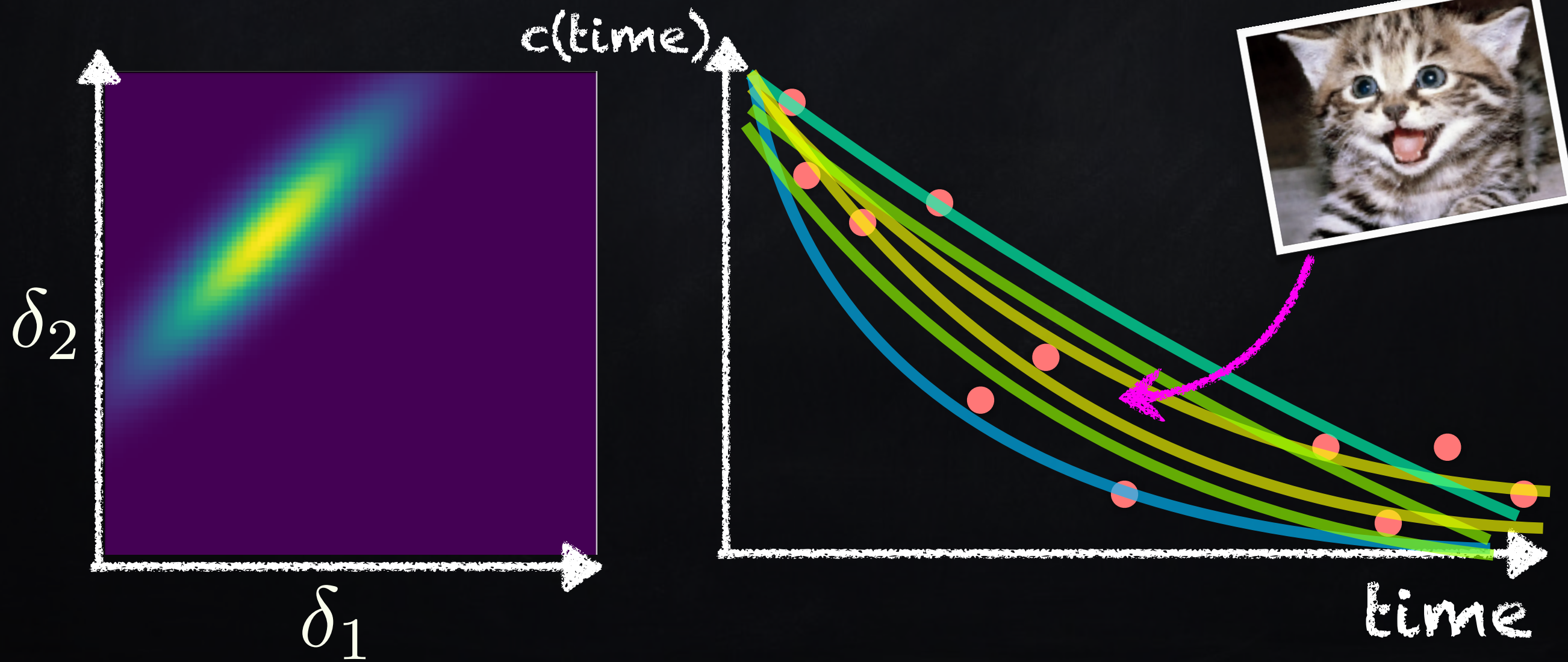
What am I supposed to do with the posterior? ✓

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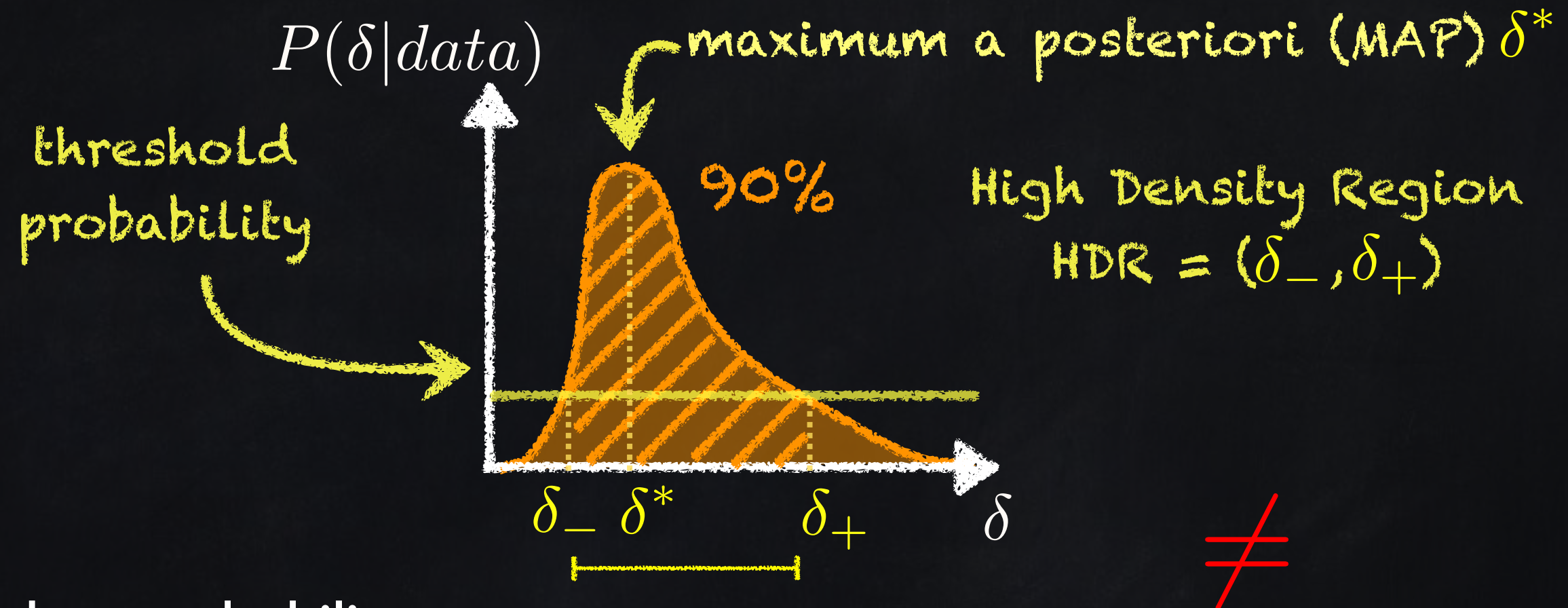
Posterior predictive checks

Correct priors?

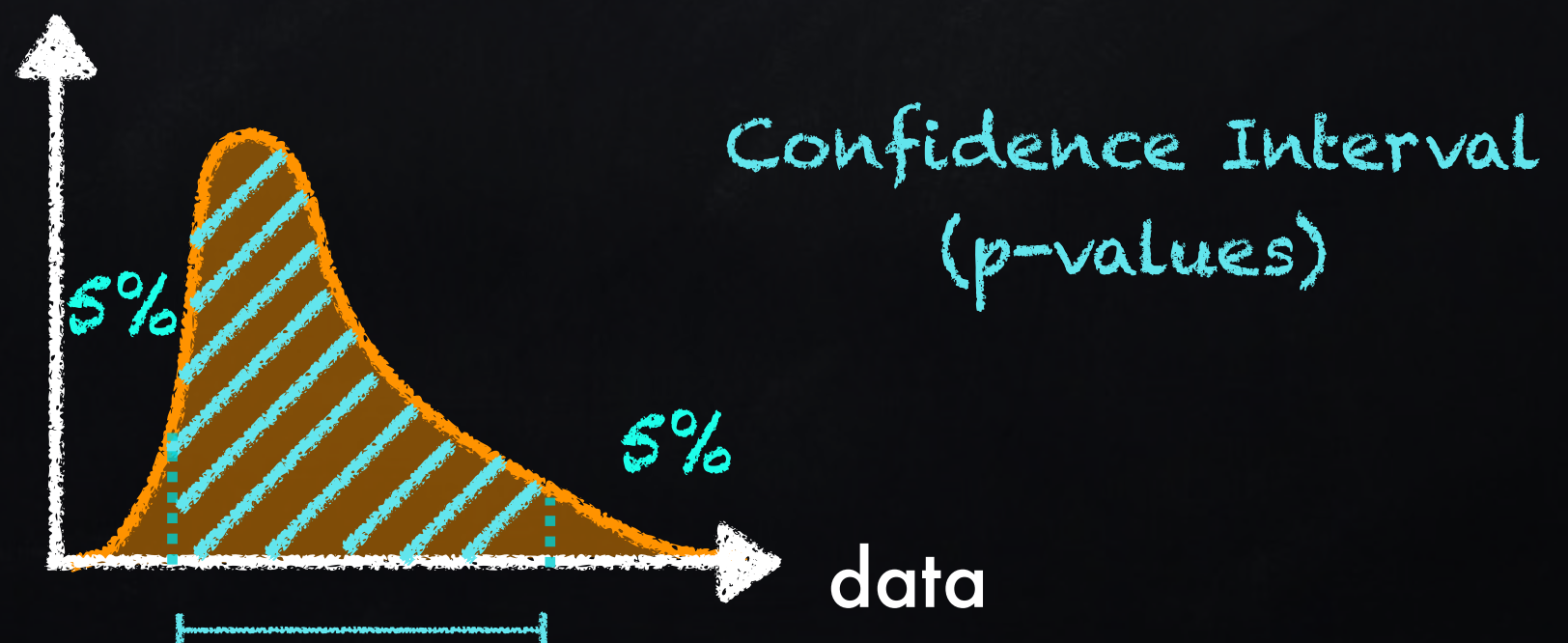
Correct likelihood/model?



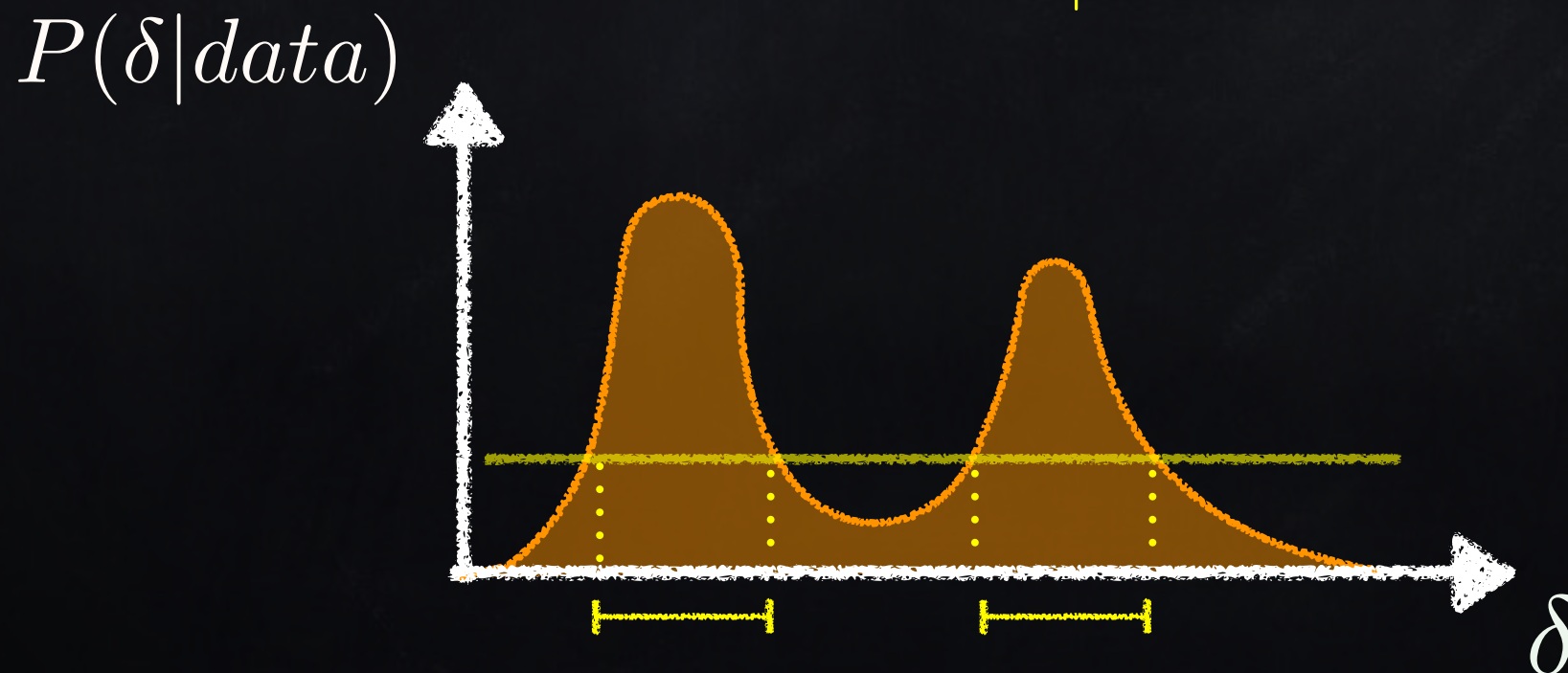
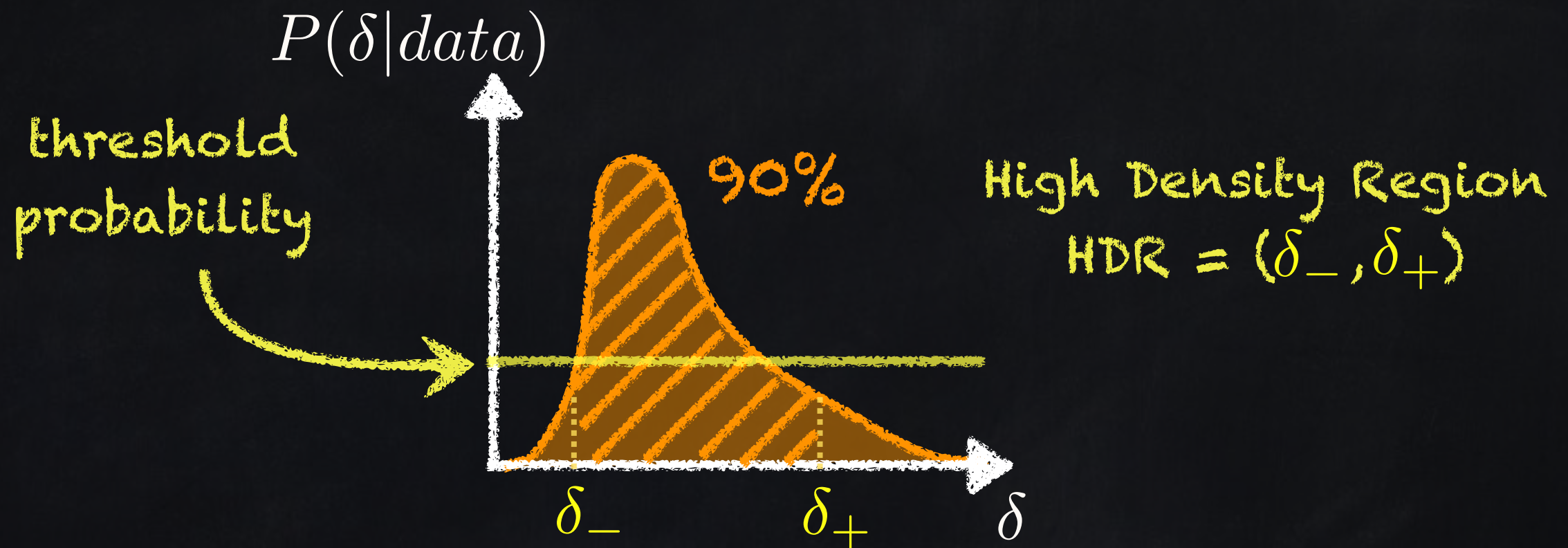
What am I supposed to do with the posterior? ✓



data probability
according
to null hypothesis

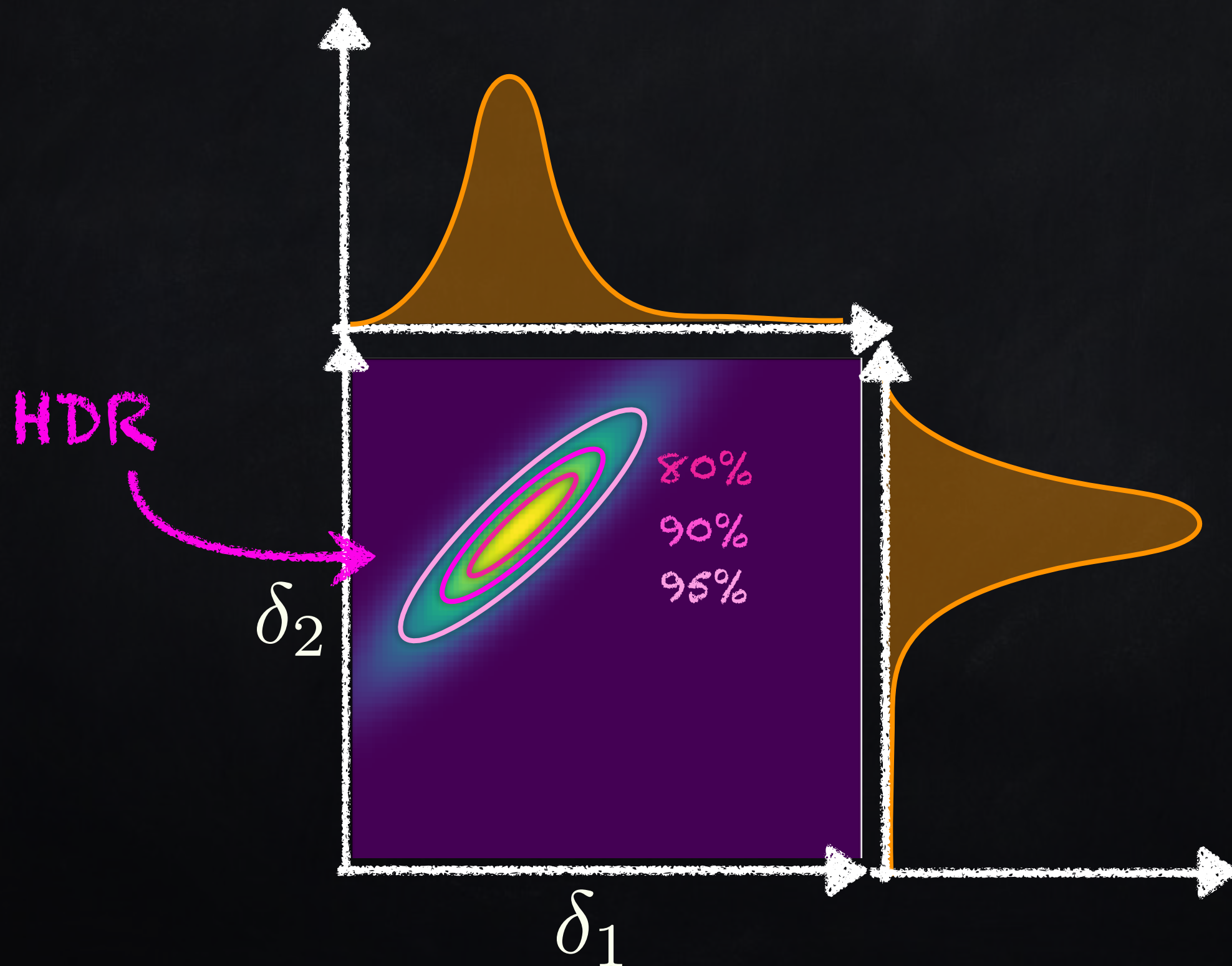


What am I supposed to do with the posterior? ✓



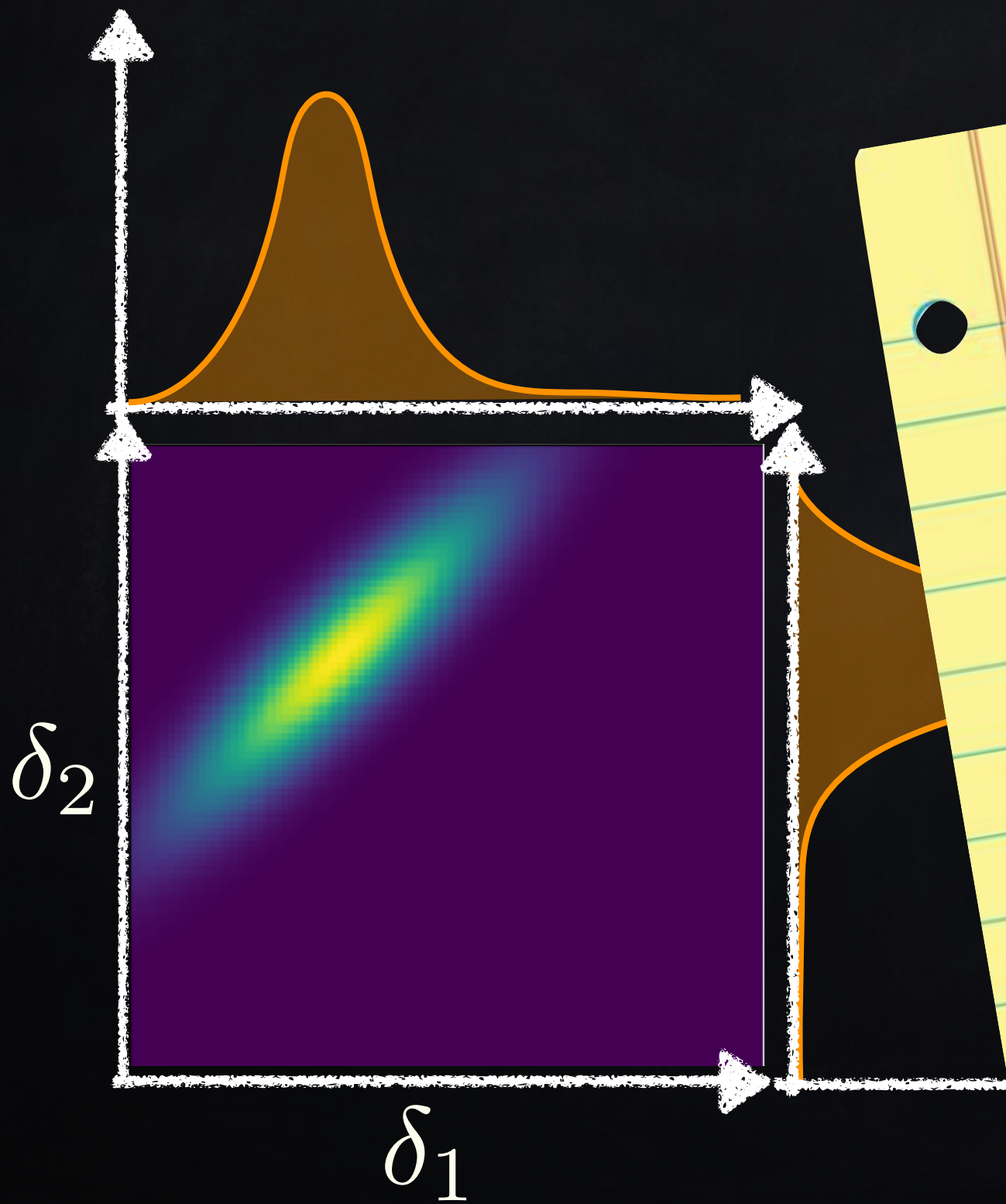
What am I supposed to do with the posterior? ✓

Inference of 2 parameters $\longrightarrow P(\delta_1, \delta_2 | data)$



What am I supposed to do with the posterior? ✓

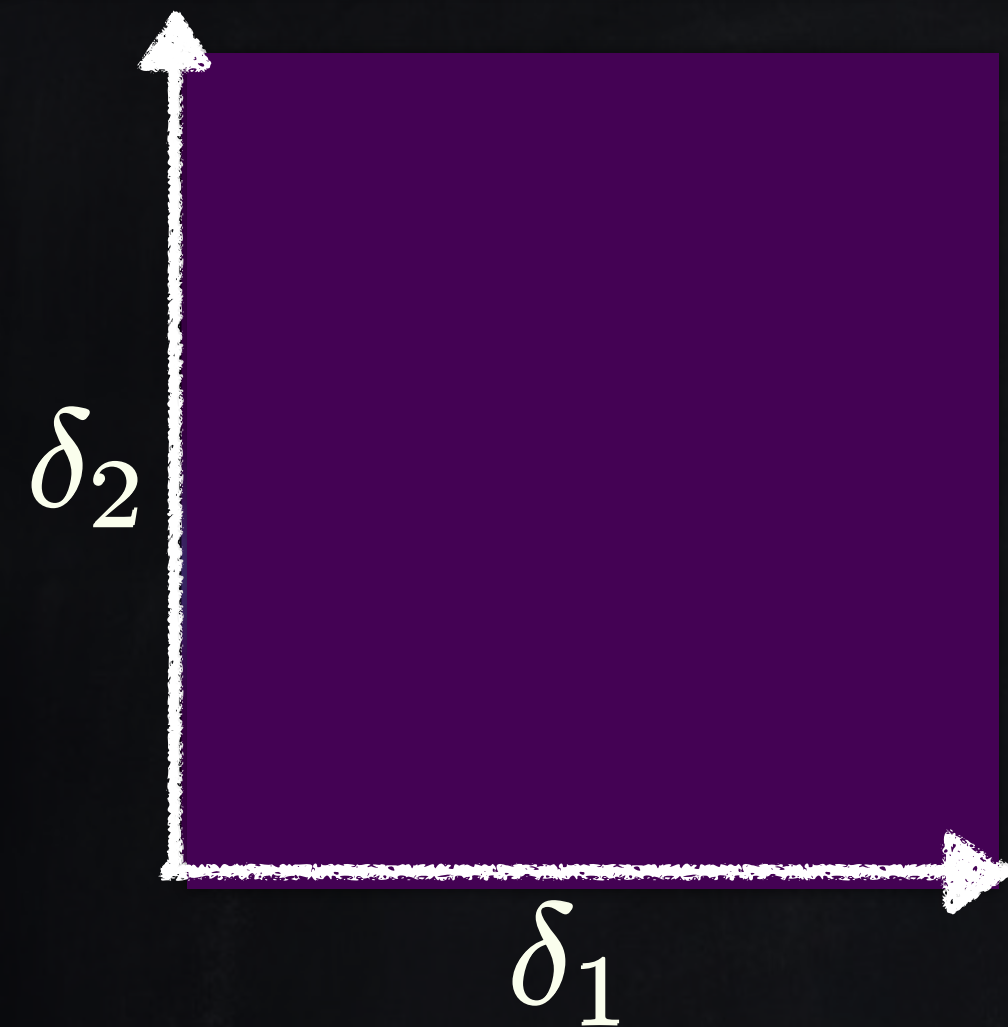
Inference of 2 parameters $\longrightarrow P(\delta_1, \delta_2 | \text{data})$



Question:
Hypothesis testing: How would you analyze the probabilities of the following hypotheses?

- $\delta_2 > A, A \in \mathbb{R}$
 $\delta_2 > \delta_1$
 $\delta_2 = \delta_1$

How the algorithms behind them work?

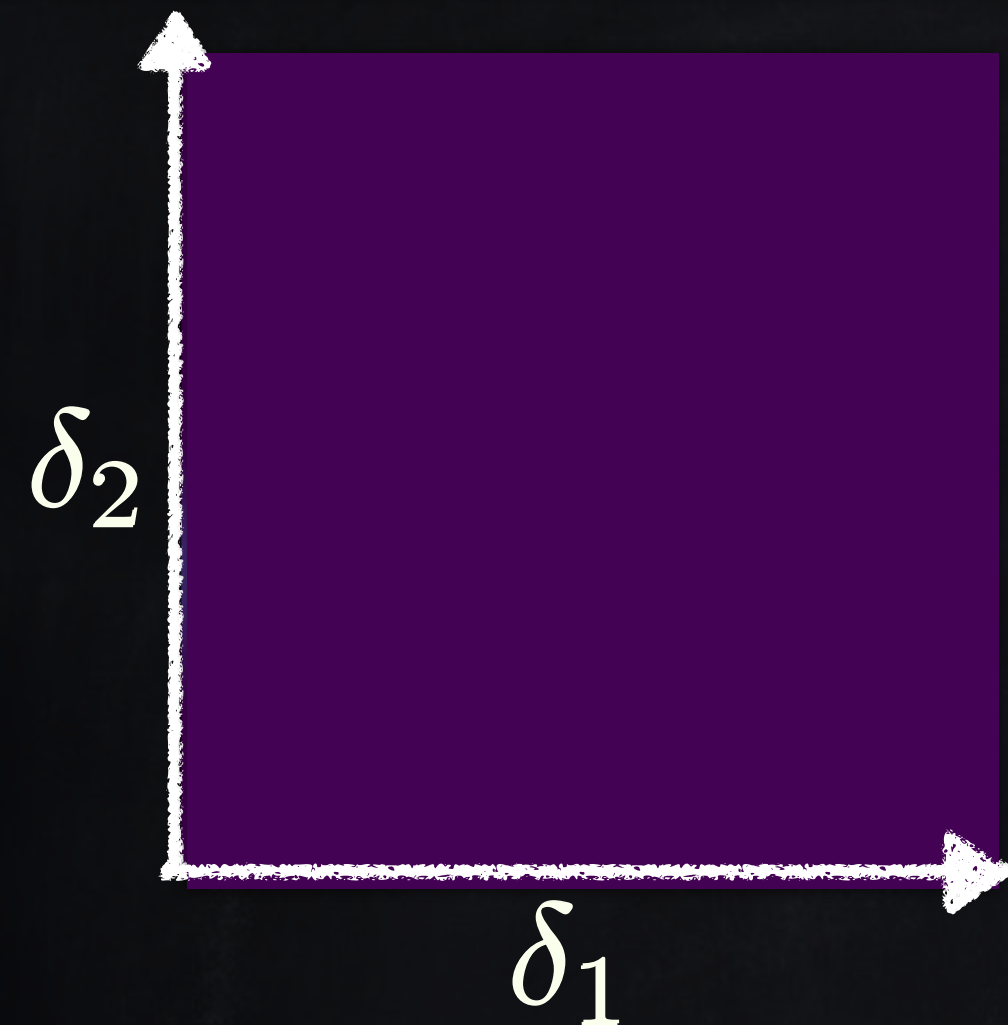


$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

Random sampling !

Markov Chain Monte Carlo (MCMC)

How the algorithms behind them work?



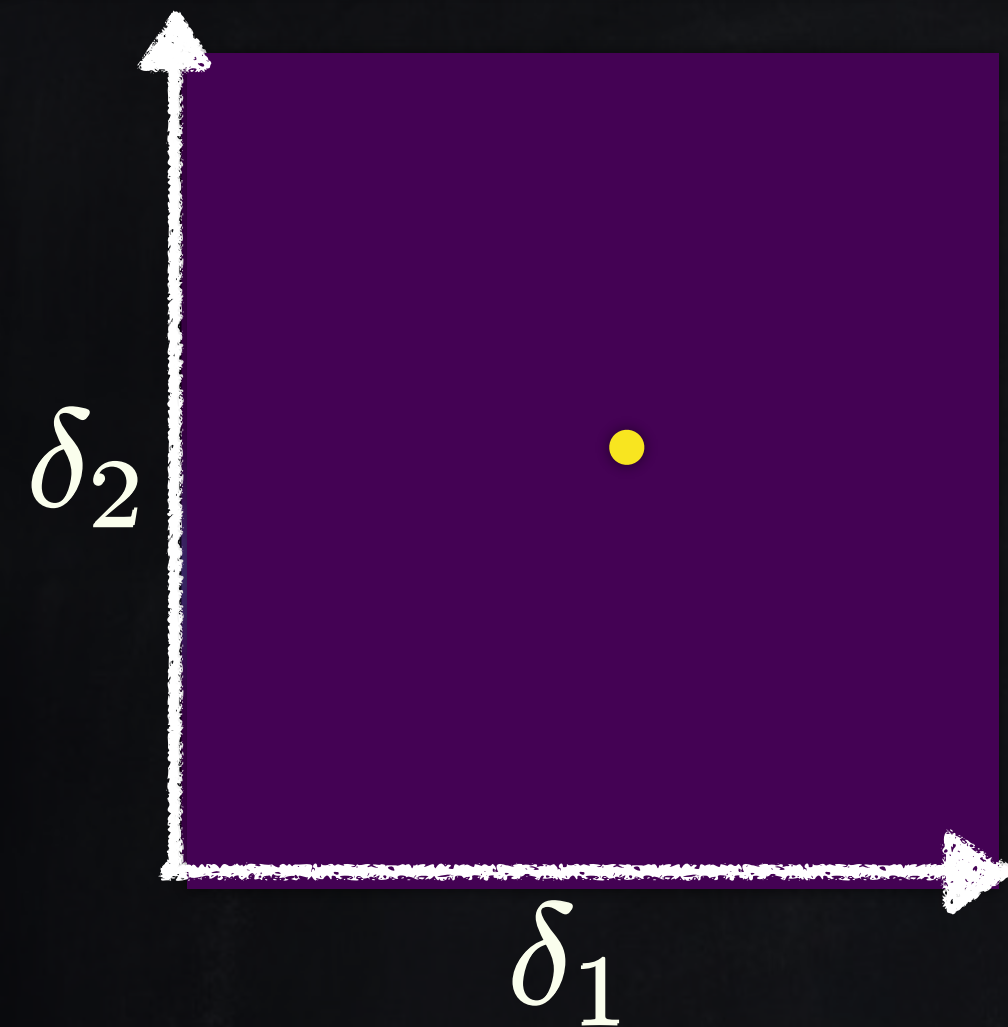
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1. Set counter for accepted points $n = 0$

How the algorithms behind them work?



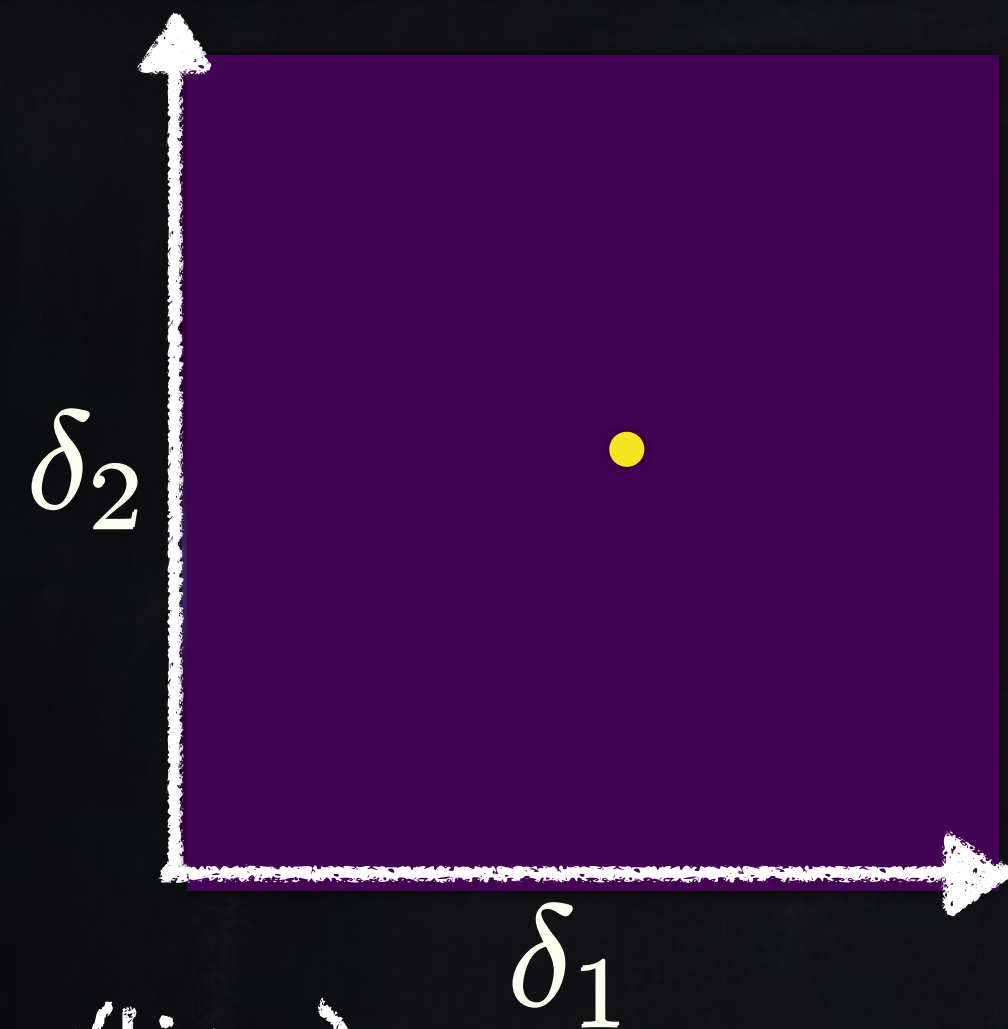
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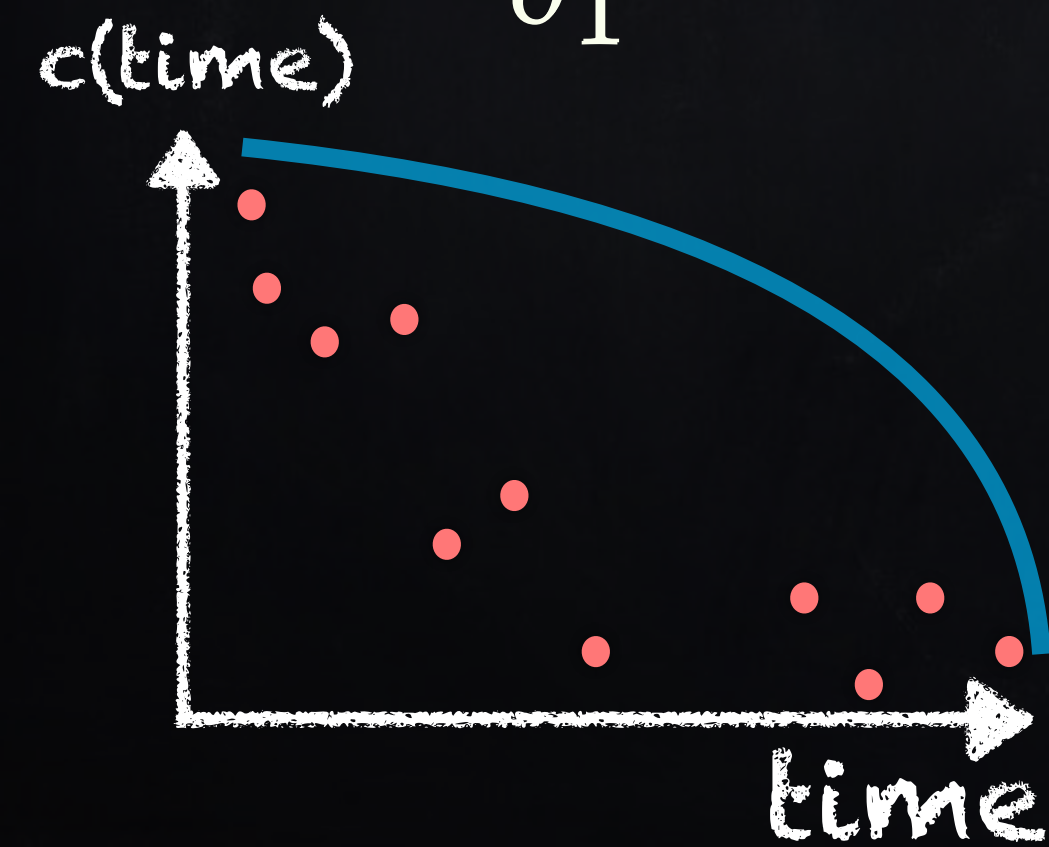


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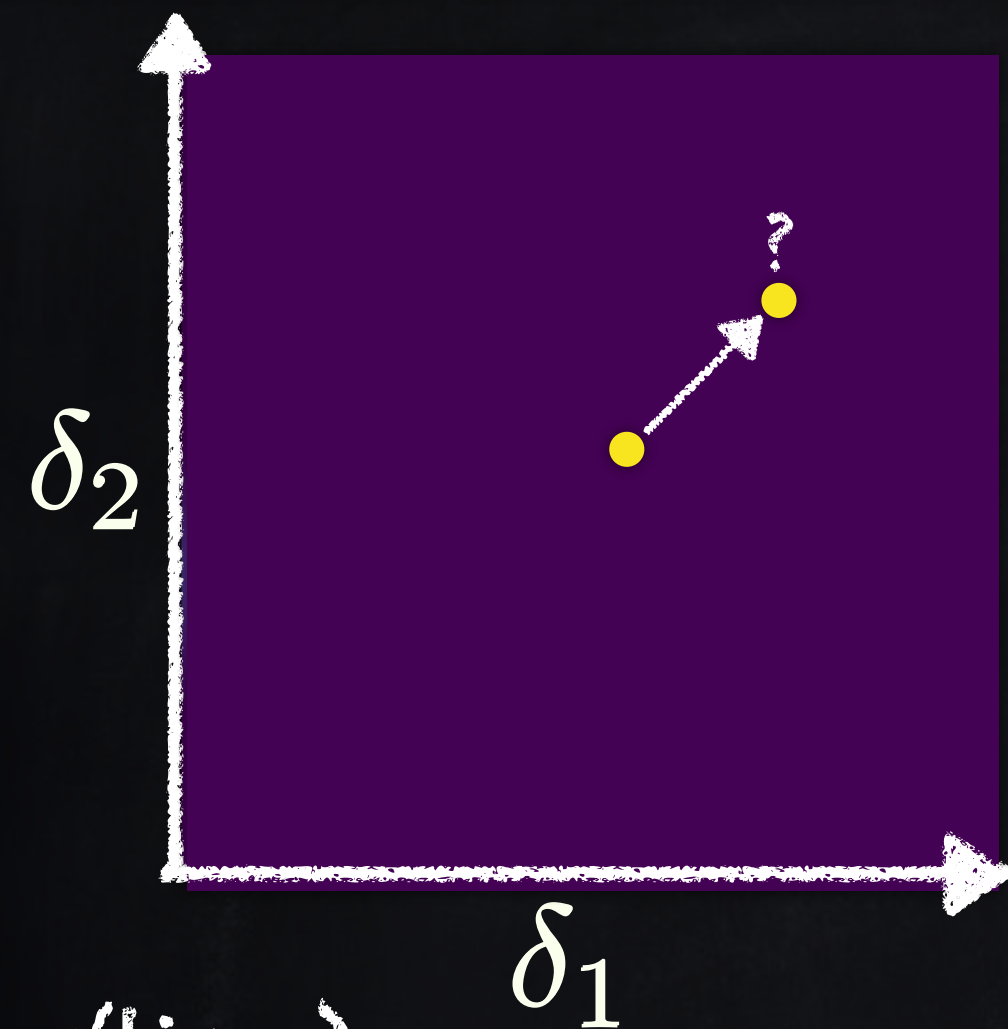
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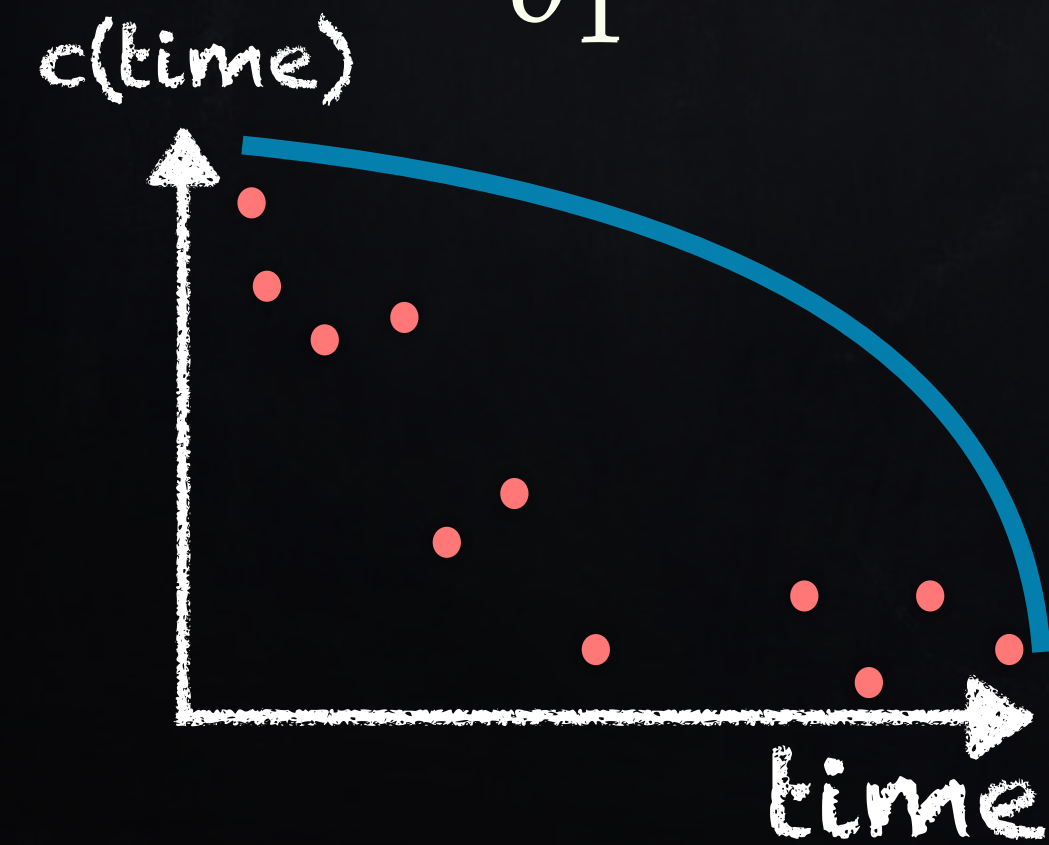


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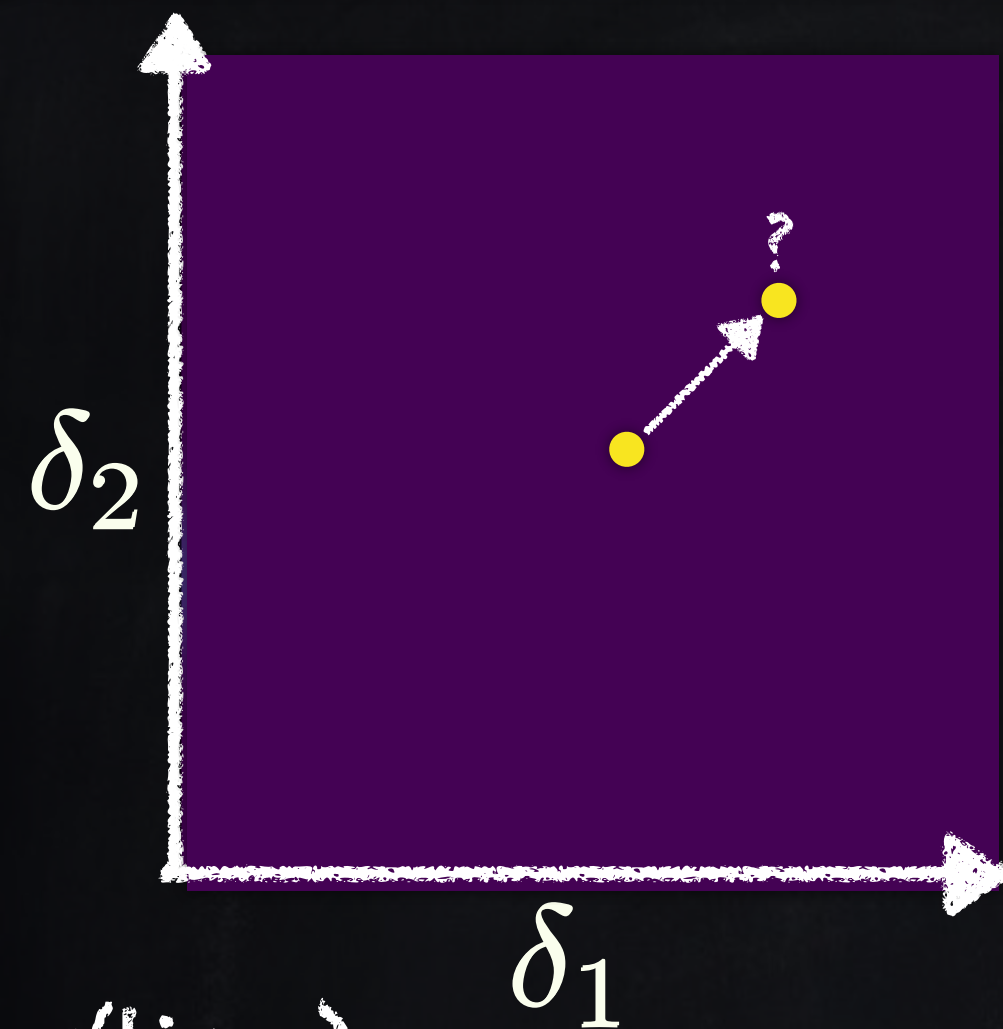
Random sampling !

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1. Set counter for accepted points $n = 0$
2. Select initial parameter point $\delta^{(0)} = (\delta_1^{(0)}, \delta_2^{(0)})$ from prior $P(\delta)$
3. Propose new parameter $\delta^{(n+1)}$ using symmetric kernel $q(\delta^{(n+1)}|\delta^{(n)})$ e.g. Gaussian



How the algorithms behind them work?

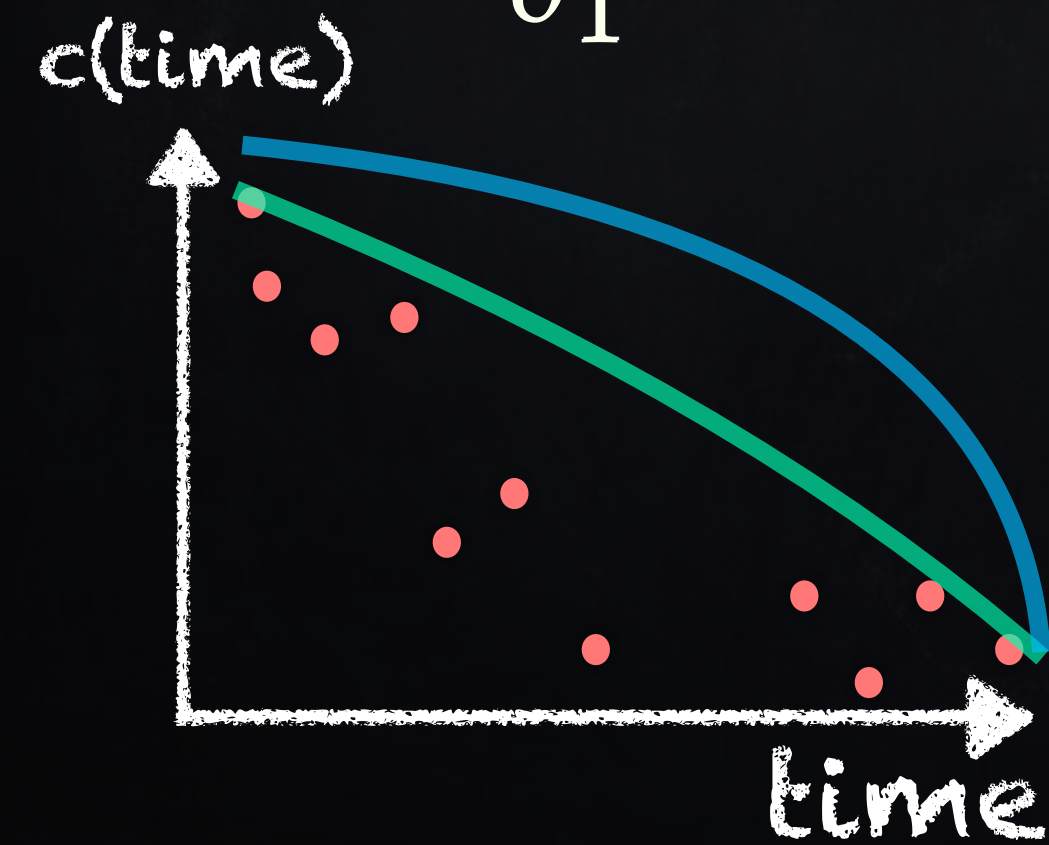


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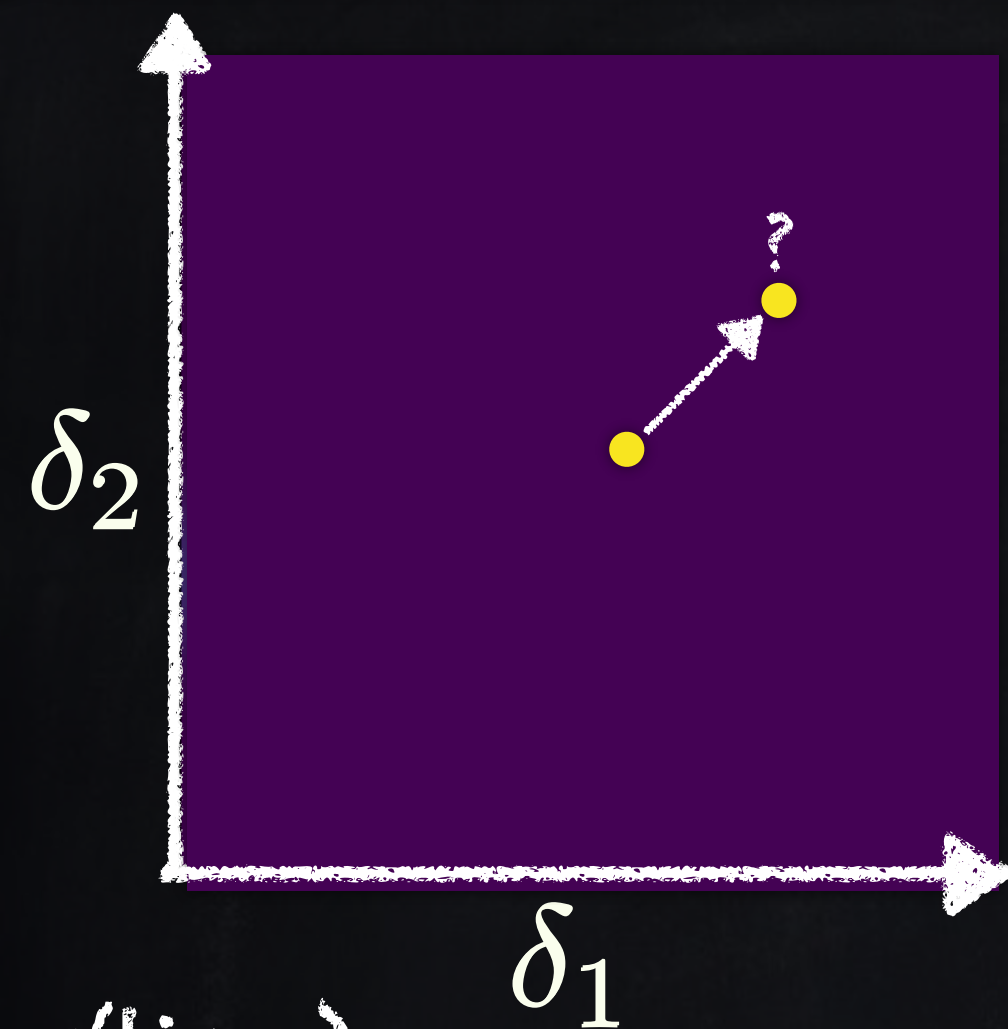
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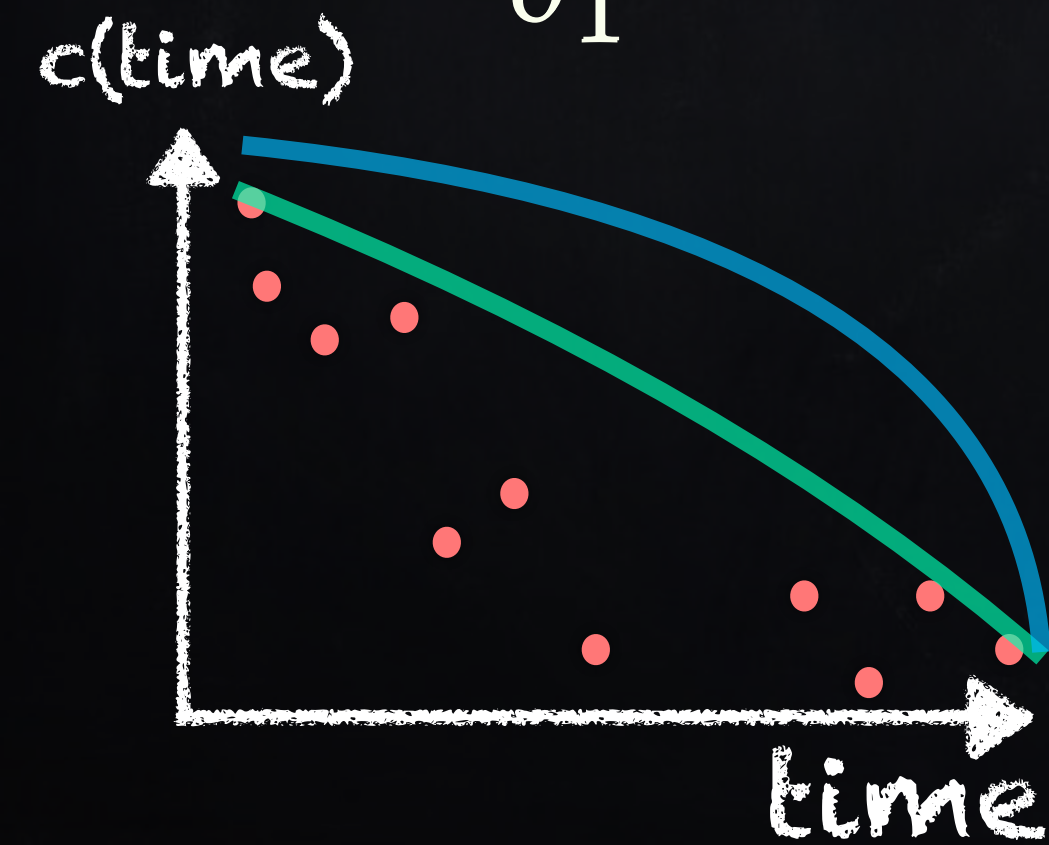
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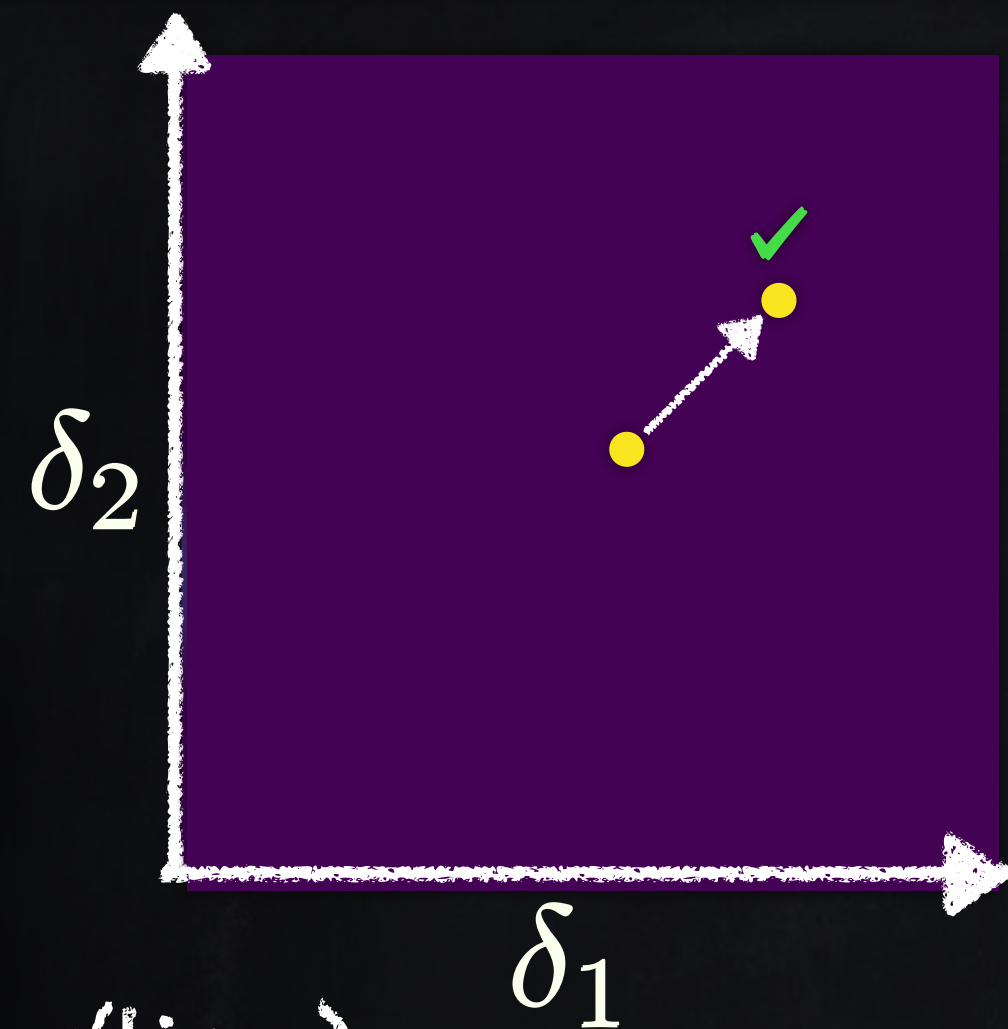
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4. Evaluate posterior ratio

$$r = \frac{P(\delta^{(n+1)}|data)}{P(\delta^{(n)}|data)} = \frac{P(data|\delta^{(n+1)})P(\delta^{(n+1)})}{P(data|\delta^{(n)})P(\delta^{(n)})}$$



How the algorithms behind them work?



$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

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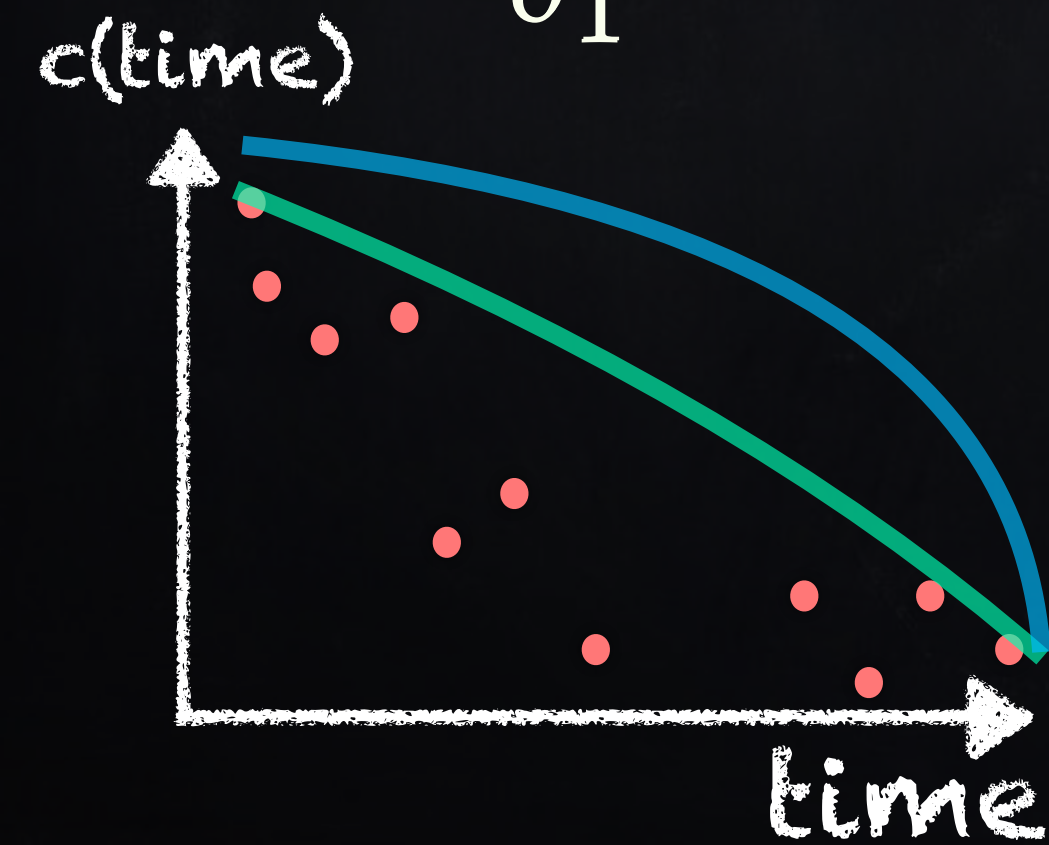
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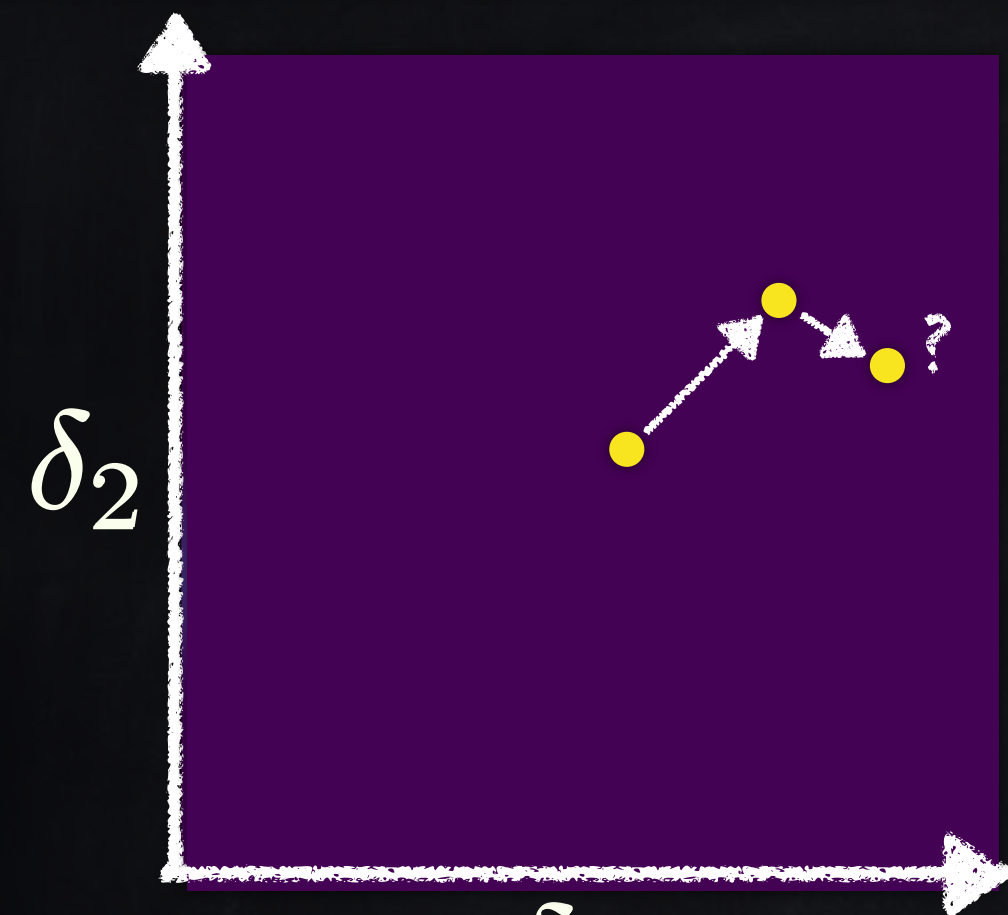
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5. Accept point with probability $\min(1, r)$



How the algorithms behind them work?



$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

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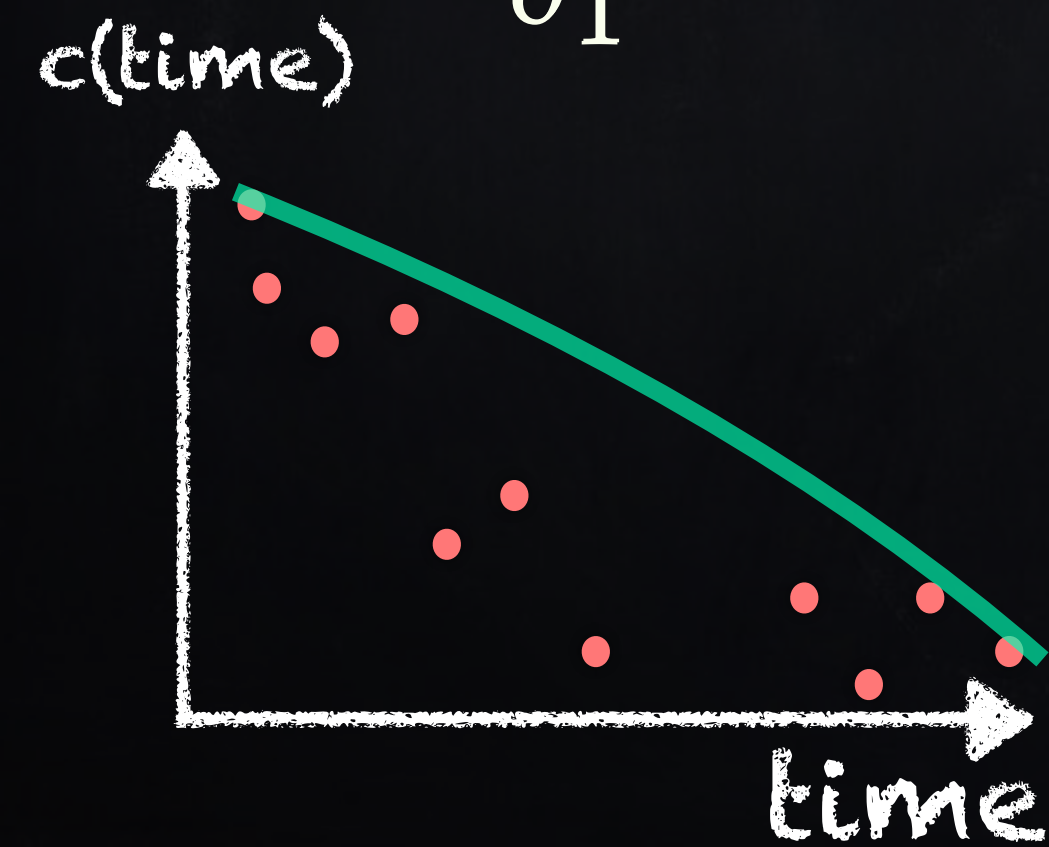
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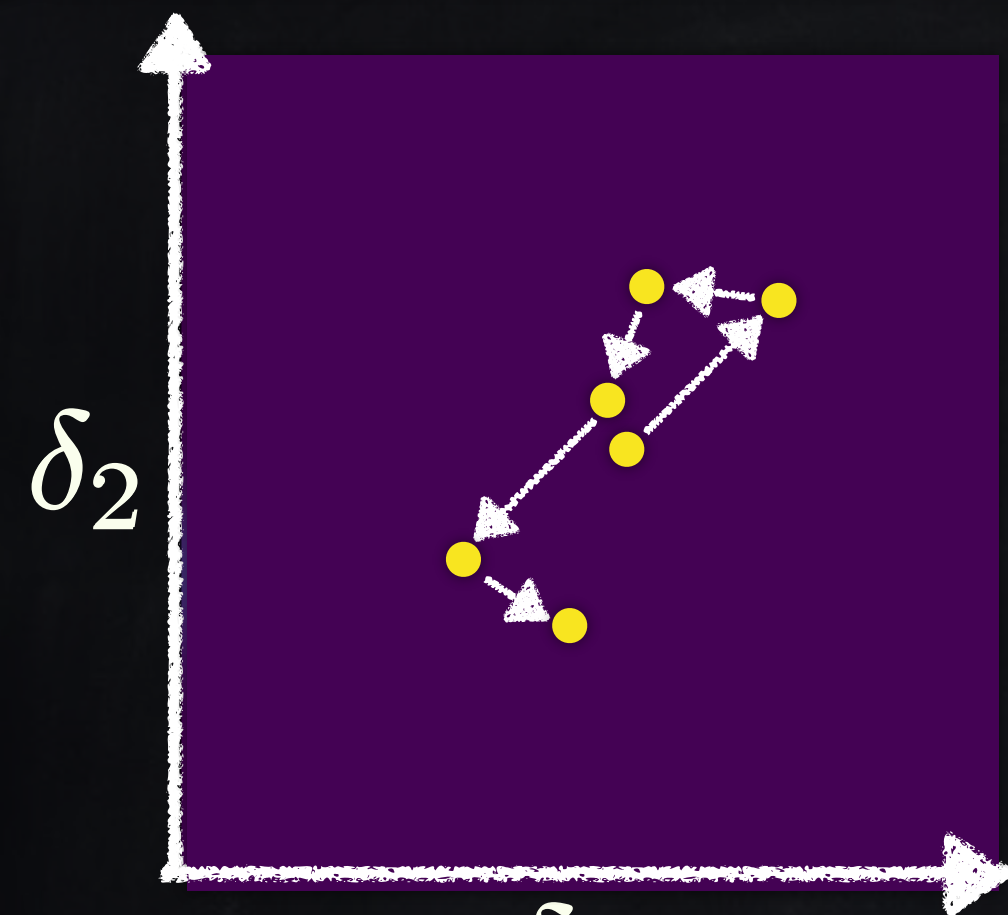
5. Accept point with probability $\min(1, r)$

6. If successful acceptance: $n = n + 1$

7. Propose next point



How the algorithms behind them work?

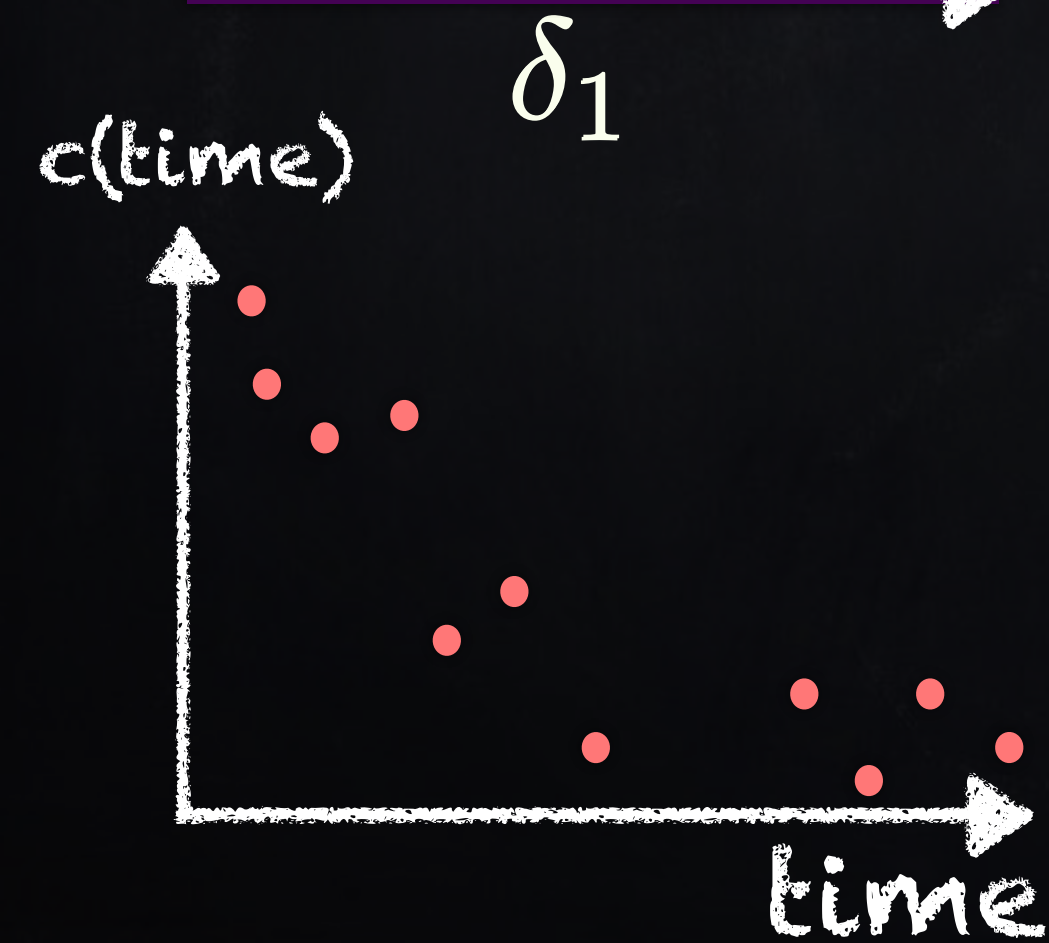


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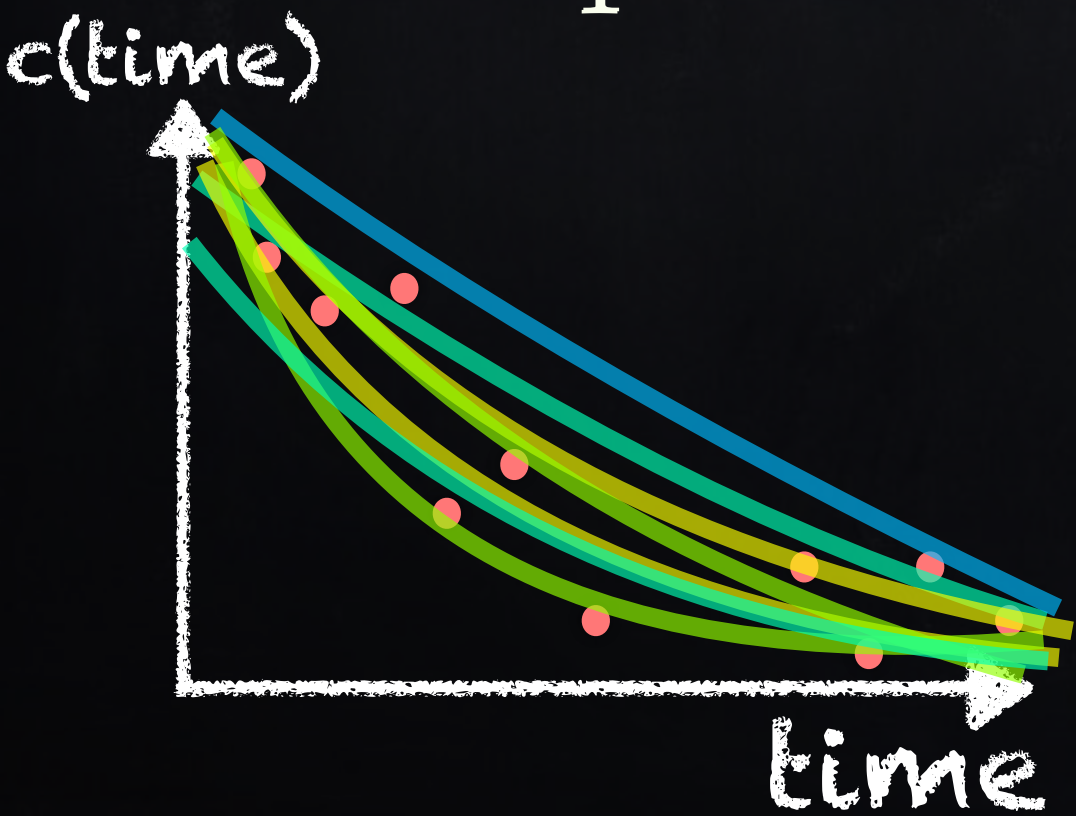
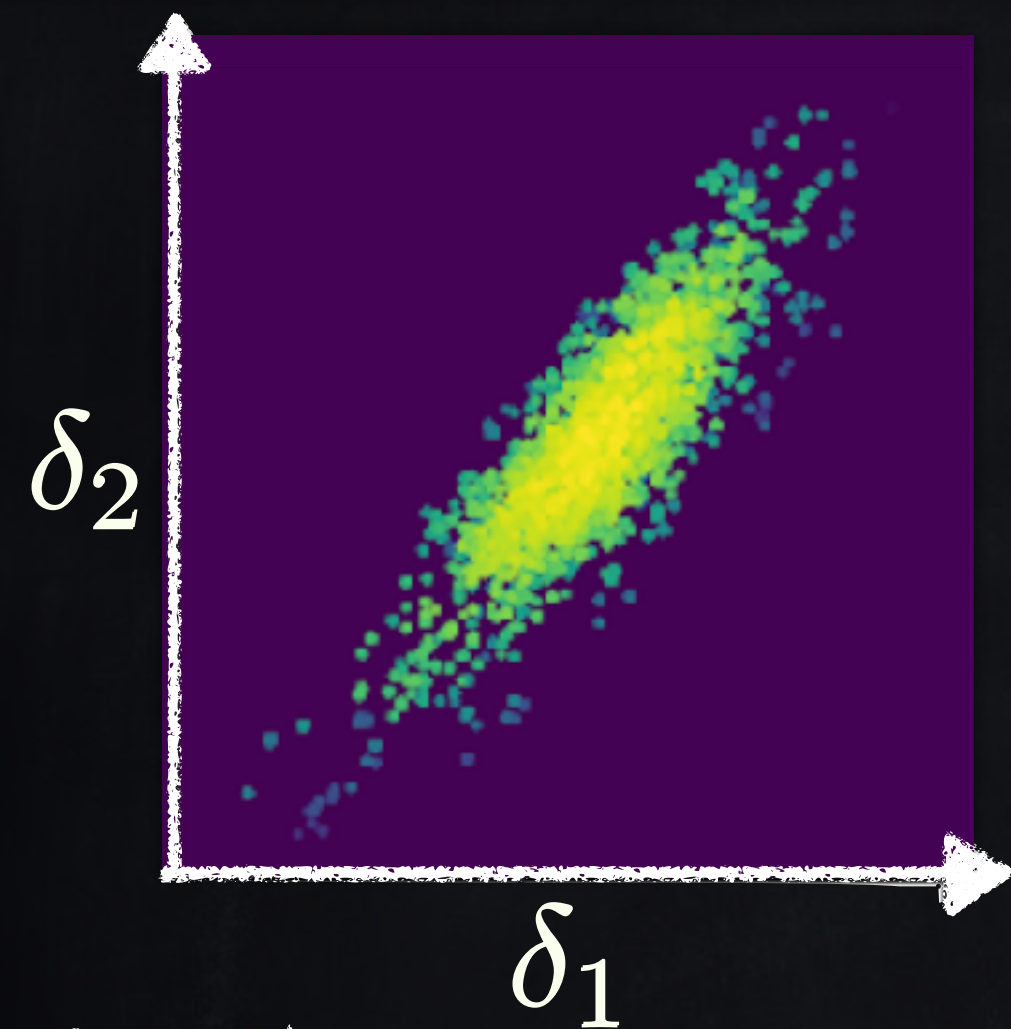
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How the algorithms behind them work?

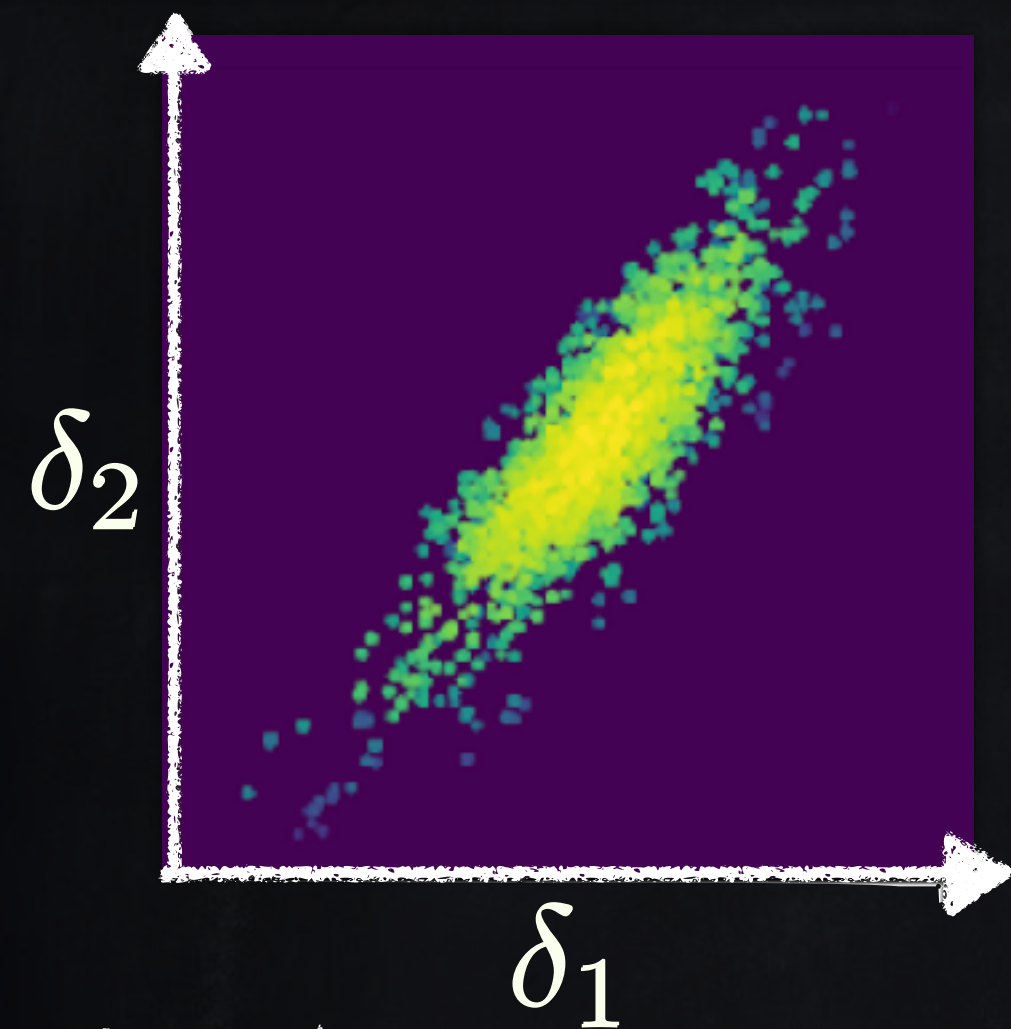


Markov Chain Monte Carlo (MCMC)

$$\delta^{(1)}, \delta^{(2)}, \delta^{(3)}, \delta^{(4)}, \dots \sim P(\delta | \text{data})$$

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5. Accept point with probability $\min(1, r)$
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How the algorithms behind them work?

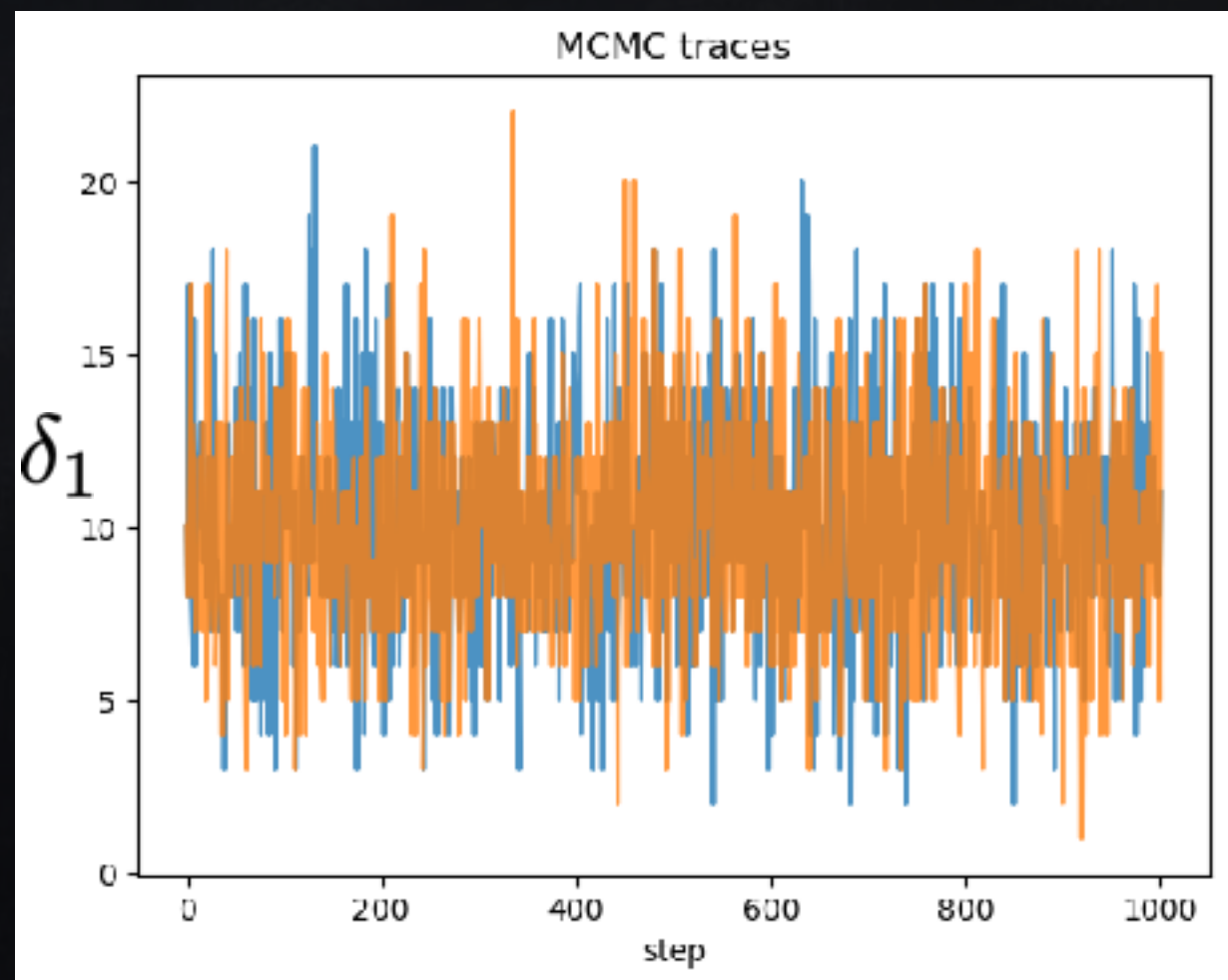
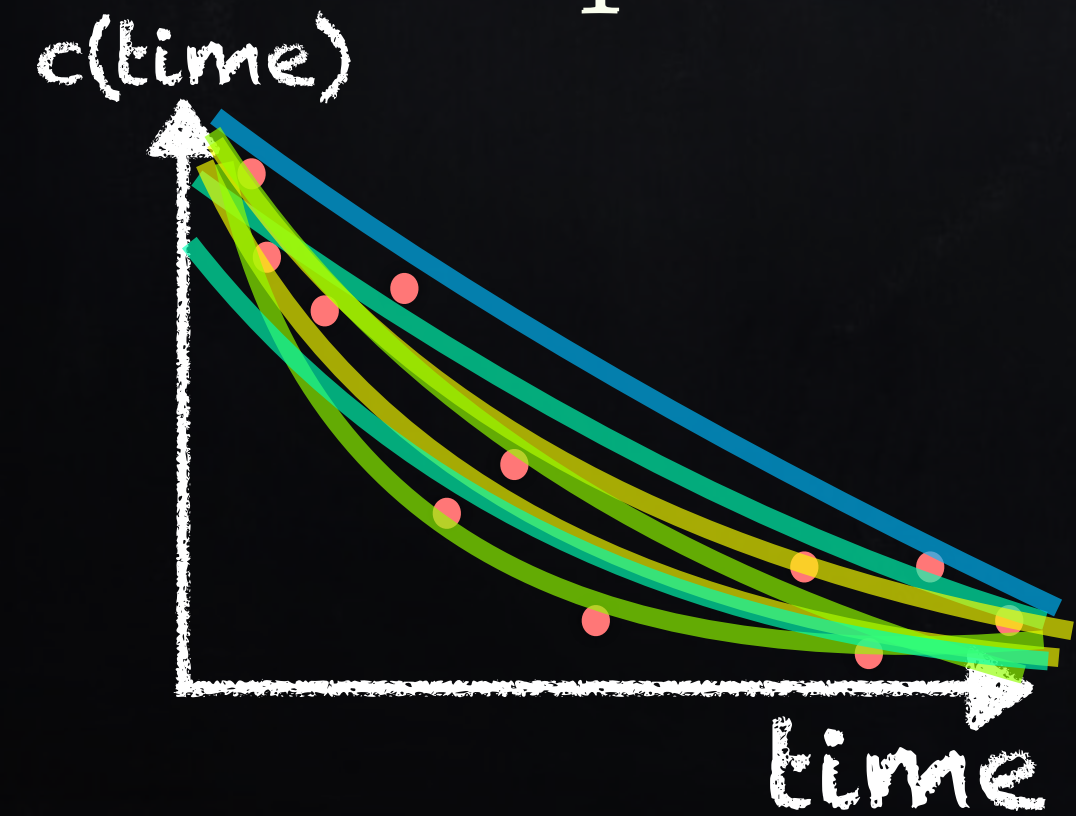


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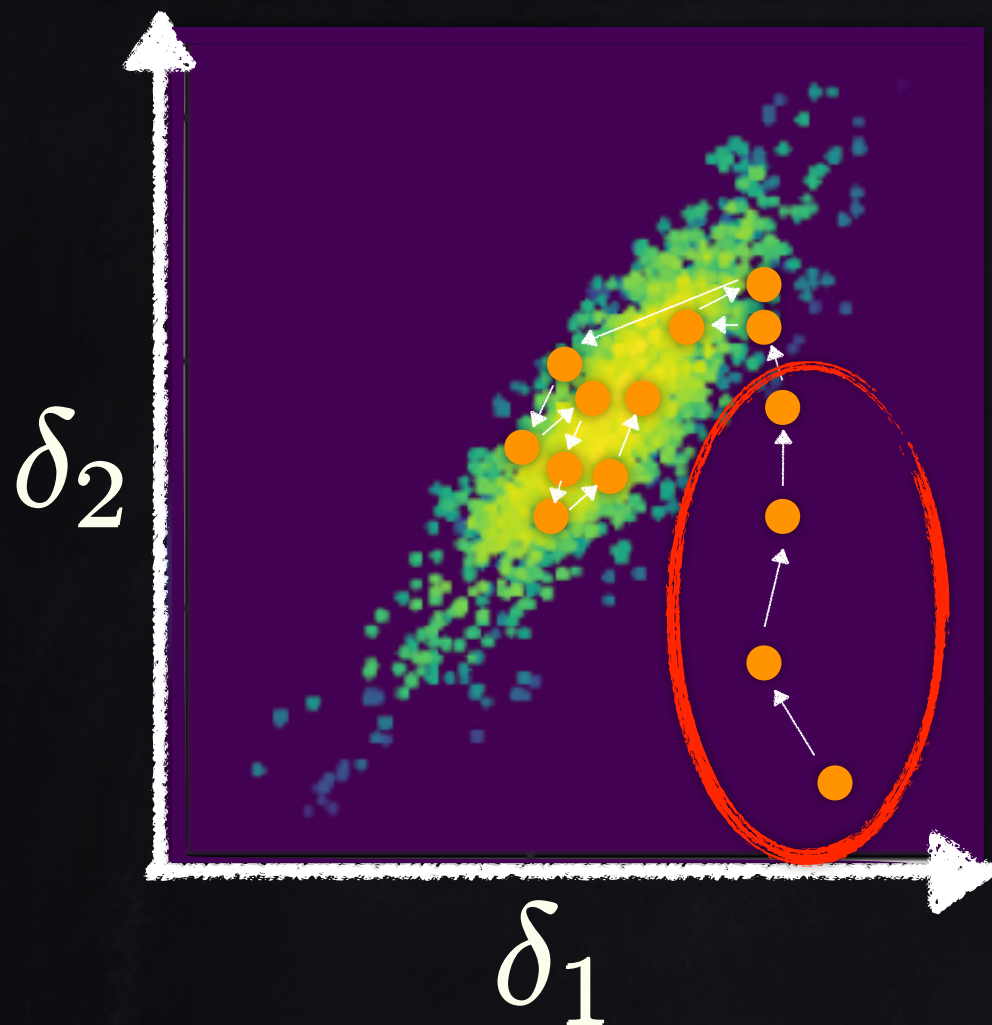
$$\delta^{(1)}, \delta^{(2)}, \delta^{(3)}, \delta^{(4)}, \dots \sim P(\delta | data)$$

MCMC diagnosis

Trace plot : plot of Markov chain progression



How the algorithms behind them work?



Markov Chain Monte Carlo (MCMC)

$$\delta^{(1)}, \delta^{(2)}, \delta^{(3)}, \delta^{(4)}, \dots \sim P(\delta|data)$$

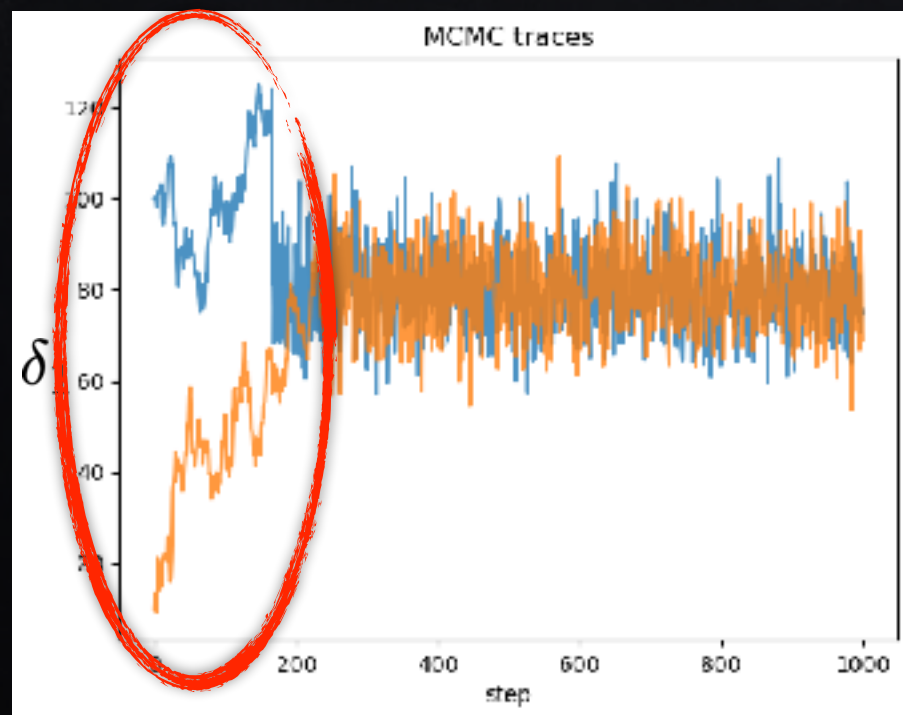
MCMC diagnosis

Case 1: “Burn-in”

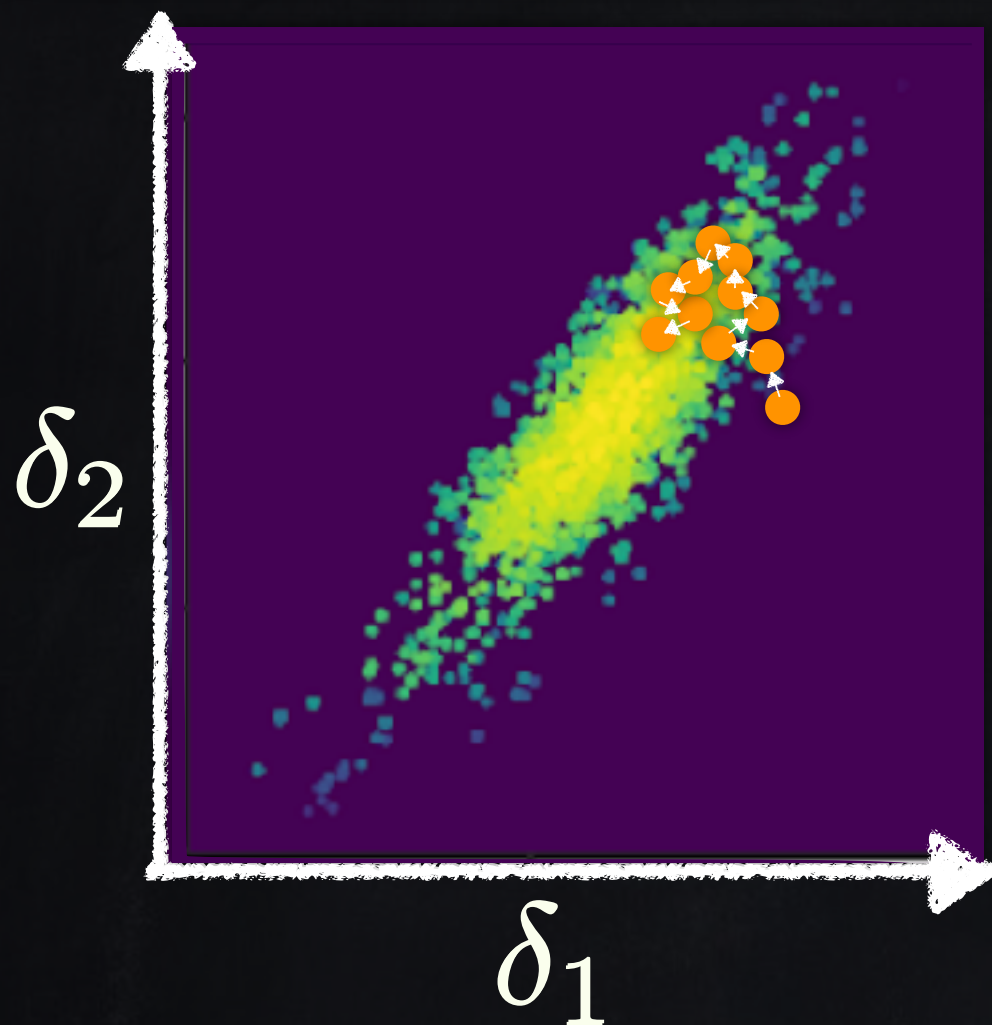
Problem: Initial part of trace not representative of $P(\delta|data)$

Solution: Remove an initial fraction of points from the MCMC

Objective measure: Variance should be constant along the chain



How the algorithms behind them work?



Markov Chain Monte Carlo (MCMC)

$$\delta^{(1)}, \delta^{(2)}, \delta^{(3)}, \delta^{(4)}, \dots \sim P(\delta | data)$$

MCMC diagnosis

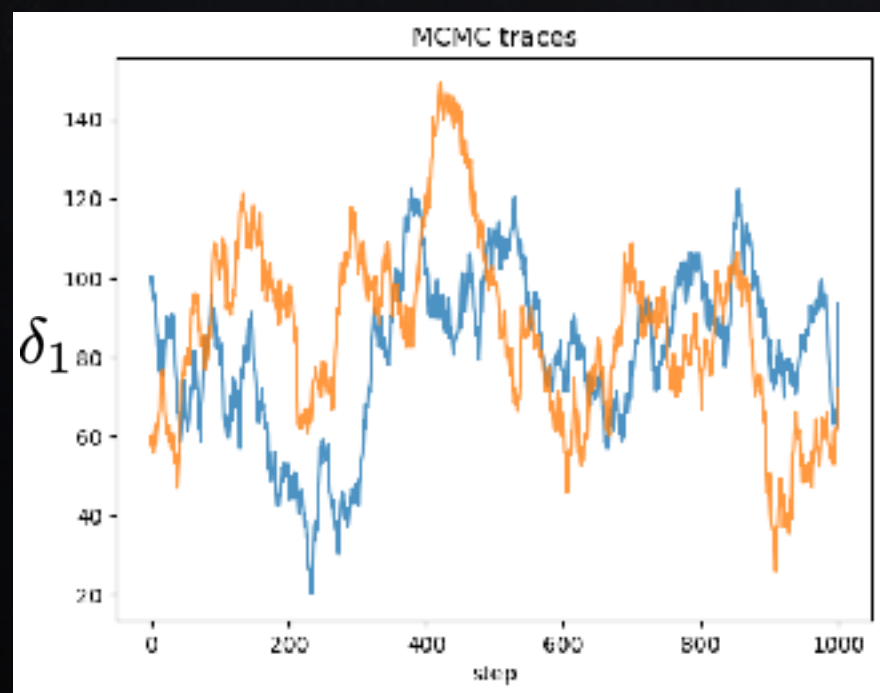
Case 2: Poor mixing

Problem: Points are correlated, limiting the information of the chain

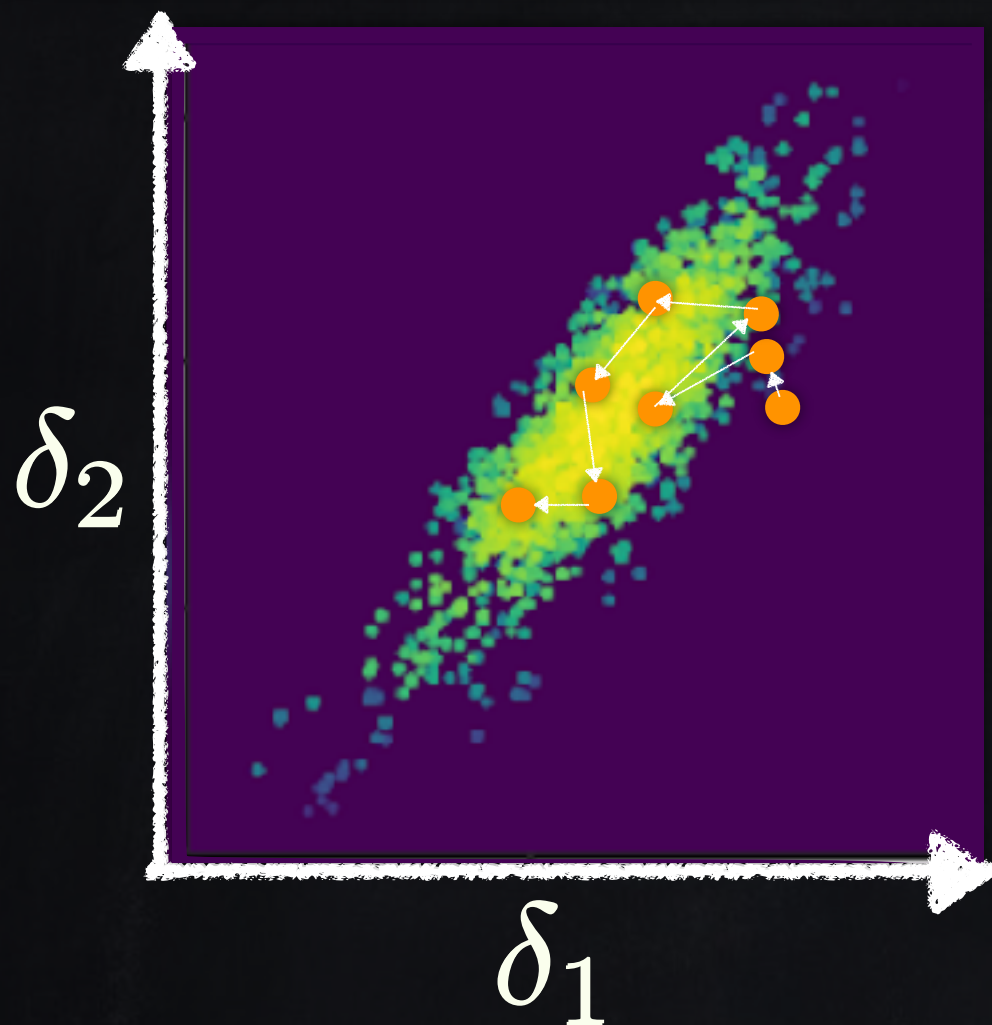
Solution: Increase size of proposal step kernel

Objective measure:
Effective sampling size

$$N_{\text{eff}} = \frac{N}{1 + 2 \sum_{\tau} \text{corr}(\delta_i, \delta_{i+\tau})} > 1000$$



How the algorithms behind them work?



Markov Chain Monte Carlo (MCMC)

$$\delta^{(1)}, \delta^{(2)}, \delta^{(3)}, \delta^{(4)}, \dots \sim P(\delta | data)$$

MCMC diagnosis

Case 3: Slow convergence

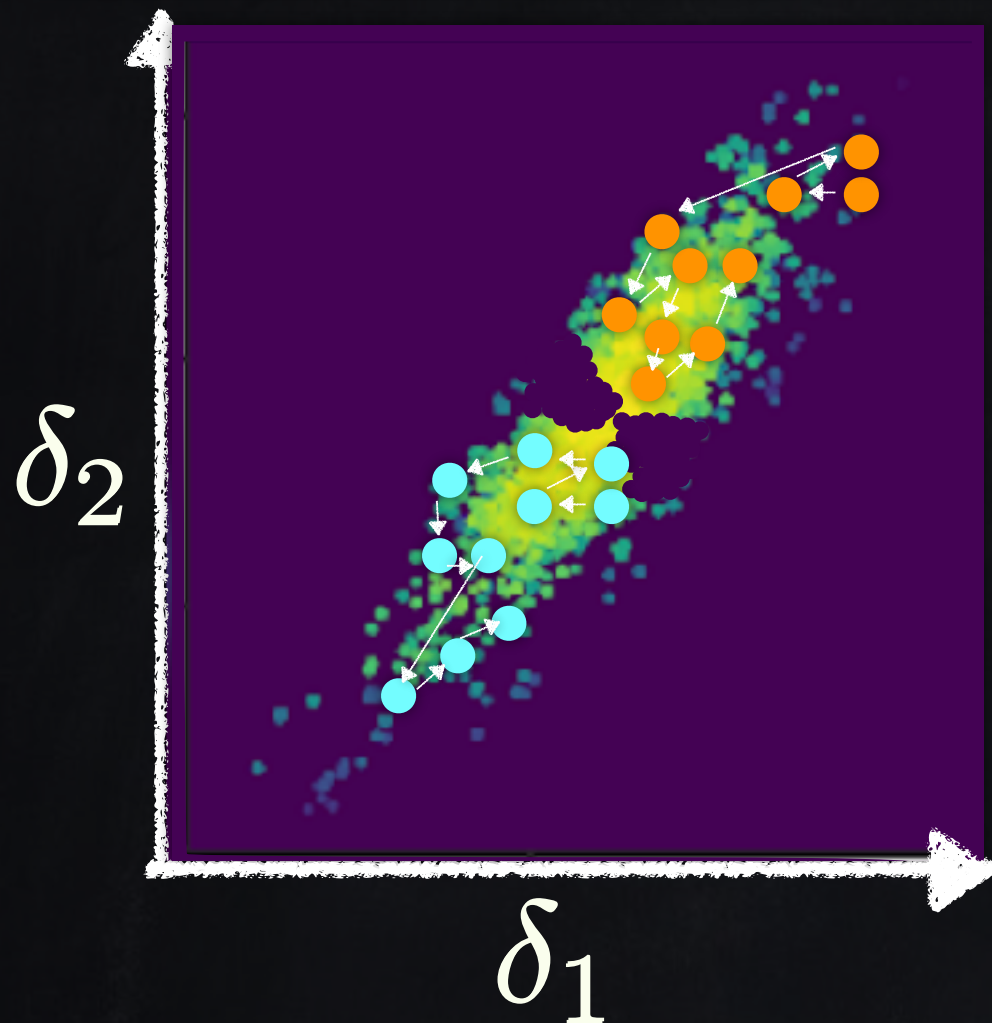
Problem: Most points proposals are rejected

Solution: Decrease size of proposal step kernel

Objective measure:

Rate of acceptance $< 10\%$

How the algorithms behind them work?



Markov Chain Monte Carlo (MCMC)

$$\delta^{(1)}, \delta^{(2)}, \delta^{(3)}, \delta^{(4)}, \dots \sim P(\delta | data)$$

MCMC diagnosis

Case 4: Multimodality /
irregular posterior shape

Problem: Different convergence across chains
Not clear steady chain convergence

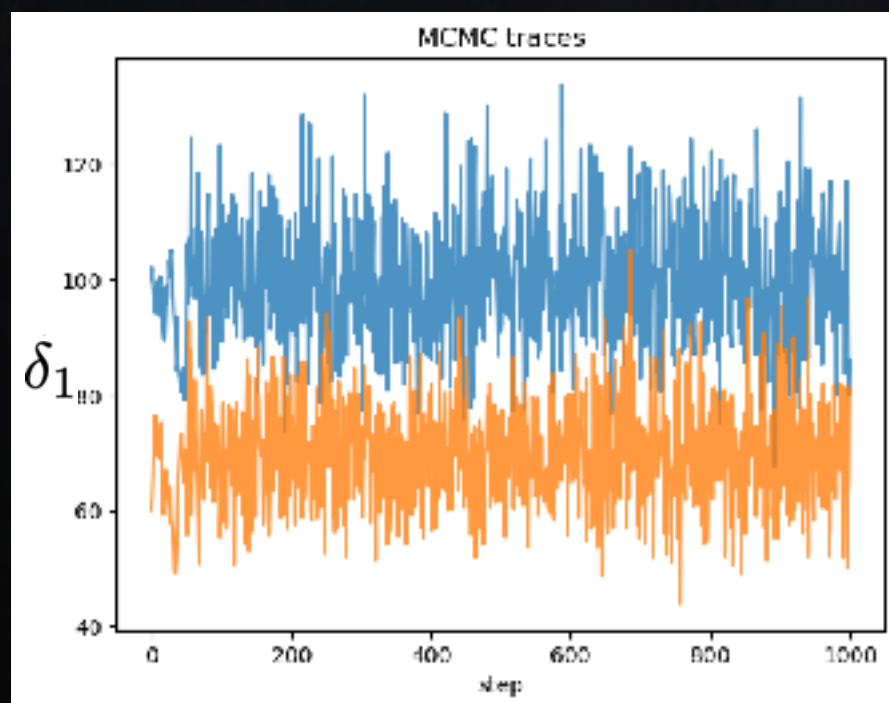
Solution: Increase chain number or chain size

Objective measure:

Gelman-Rubin statistic ($\hat{R}$)

Compares variance within chains
with variance between chains

$$1 < \hat{R} < 1.1$$



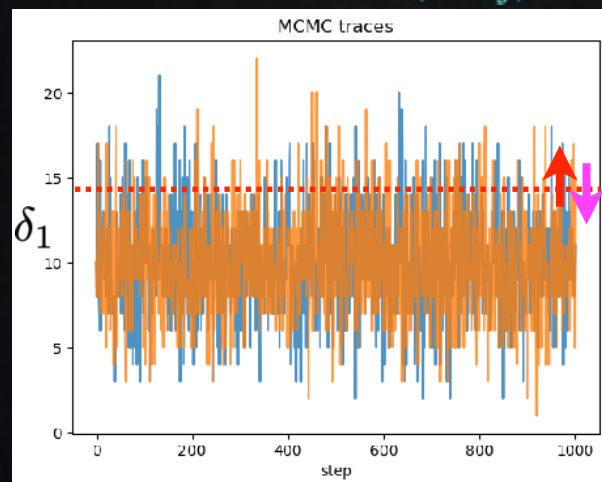
How the algorithms behind them work? ✓

OK, but WHY does MCMC work?? ✓

MCMC steady state probability ? Posterior probability

$$P^{MC}(\delta) = P(\delta)$$

$$P^{MC}(\delta^{(i)} \rightarrow \delta^{(j)}) = P^{MC}(\delta^{(j)} \rightarrow \delta^{(i)}) \\ \equiv P^{MC}(\delta_i, \delta_j)$$



$$P^{MC}(\delta^{(n)}, \delta^{(n+1)}) = P^{MC}(\delta^{(n)}) P^{MC}(\delta^{(n+1)} | \delta^{(n)})$$

MCMC algorithm

$$P^{MC}(\delta^{(n)}, \delta^{(n+1)}) = P^{MC}(\delta^{(n+1)}) P^{MC}(\delta^{(n)} | \delta^{(n+1)}) \quad \div$$

$$\frac{P^{MC}(\delta^{(n)})}{P^{MC}(\delta^{(n+1)})} = \frac{P^{MC}(\delta^{(n)} | \delta^{(n+1)})}{P^{MC}(\delta^{(n+1)} | \delta^{(n)})} = \frac{\cancel{q(\delta^{(n)}, \delta^{(n+1)})} \min(1, P(\delta^{(n)}) / P(\delta^{(n+1)}))}{\cancel{q(\delta^{(n+1)}, \delta^{(n)})} \min(1, P(\delta^{(n+1)}) / P(\delta^{(n)}))} \\ = \frac{P(\delta^{(n)})}{P(\delta^{(n+1)})} \quad \checkmark \text{ q.e.d.}$$



fun!! Can you give more examples of Likelihood? ✓

Bayesian inference methods for ecology

Theory

What is Bayesian inference?

Probability 101

Bayes' theorem in biology

Priors and likelihood

Interpretation of posteriors

Markov Chain Monte Carlo

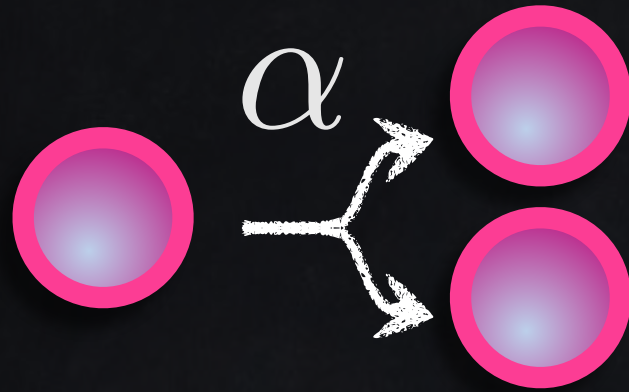
Applications

Population dynamics in
microscopic and
macroscopic communities

ODE generative models:
Morphogen patterning of
embryonic tissues

Phylogeny and model
selection

Example 1.1: Determining a proportion



probability of proliferation



$$p = \frac{\alpha}{\alpha + \beta}$$



probability of apoptosis



$$q = 1 - p = \frac{\beta}{\alpha + \beta}$$

This example applies to any experiment in which you are measuring the fraction of a trait with two possible outcomes

e.g. fraction of males in a population of penguins

fraction of trees in a forest with a specific disease

fraction of mating attempts that are successful

practical session



Example 1.1: Determining a proportion

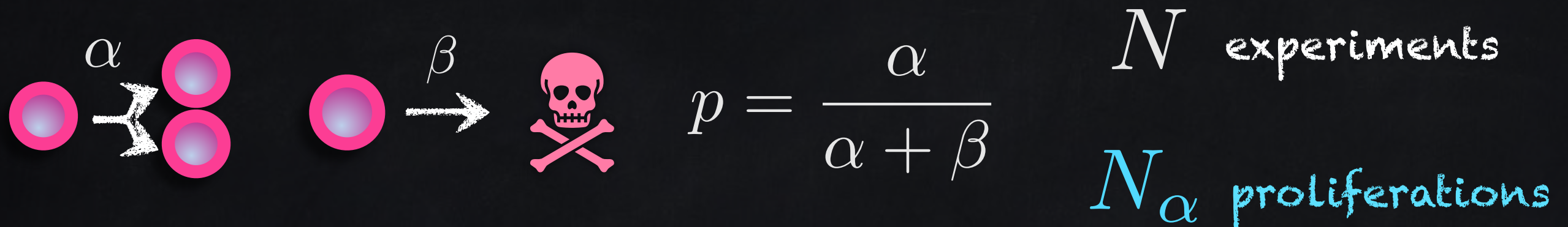


$$p = \frac{\alpha}{\alpha + \beta}$$

	Proliferated	Died
N experiments	XXXXX	XXXXX
N_α proliferations	XXXXX	
	XXXXX	
	XXXXX	
	XX	

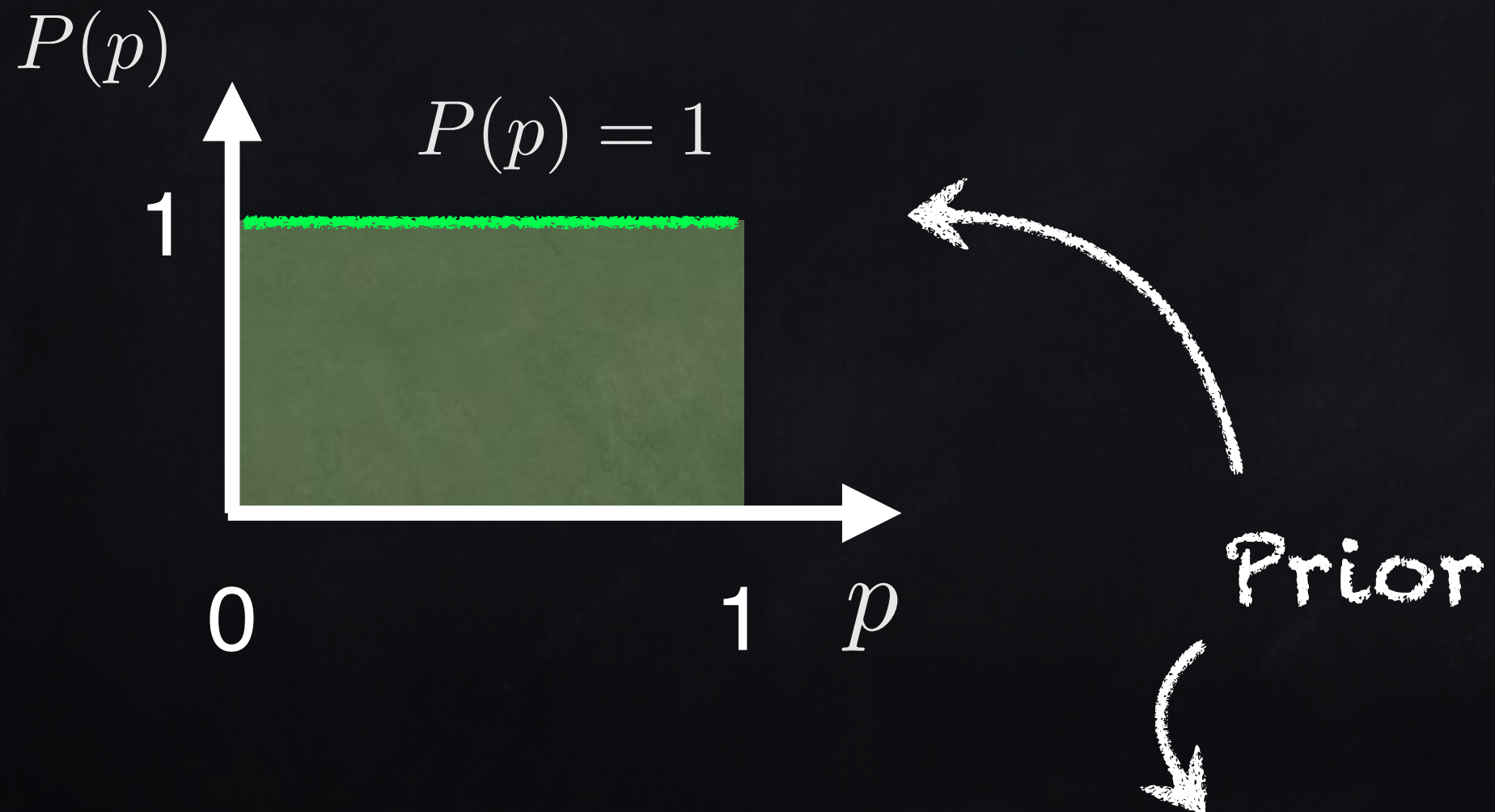
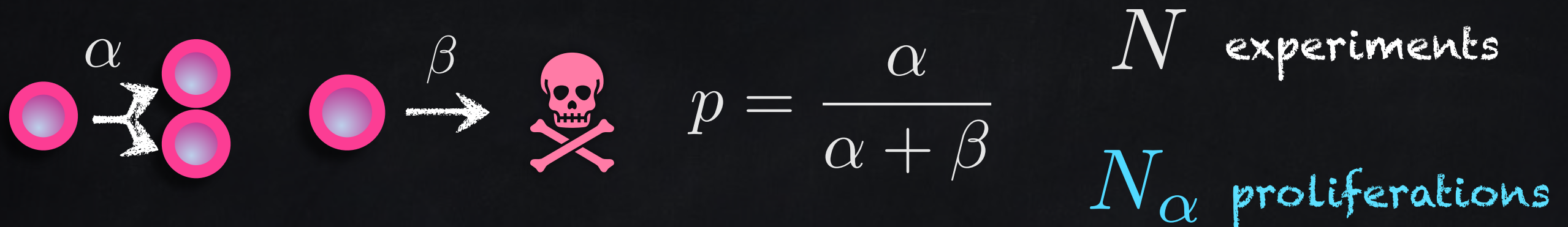
$$P(p|N_\alpha) = \frac{P(N_\alpha|p)P(p)}{P(N_\alpha)}$$

Example 1.1: Determining a proportion



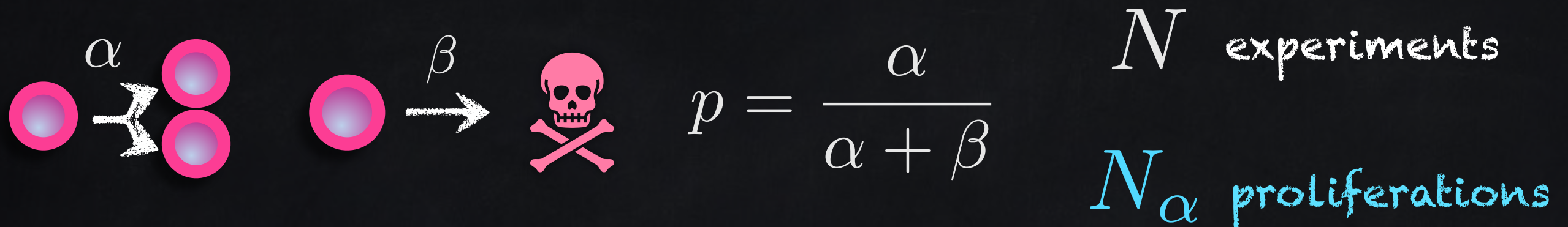
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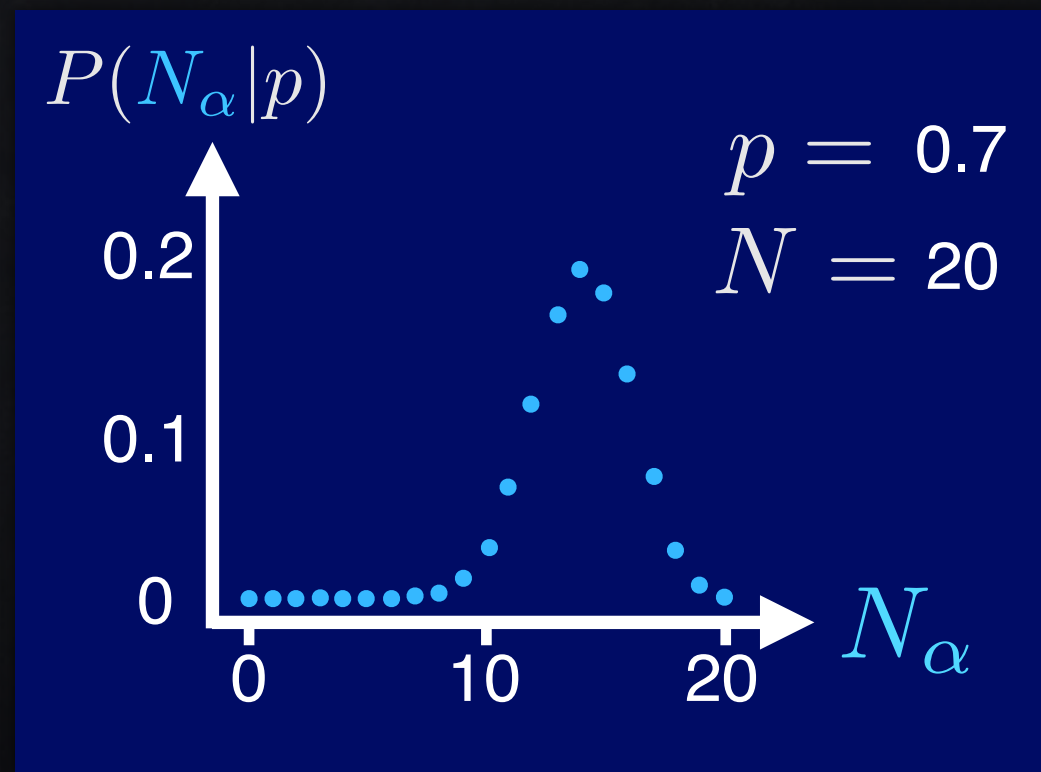
Example 1.1: Determining a proportion



Likelihood follows a binomial distribution

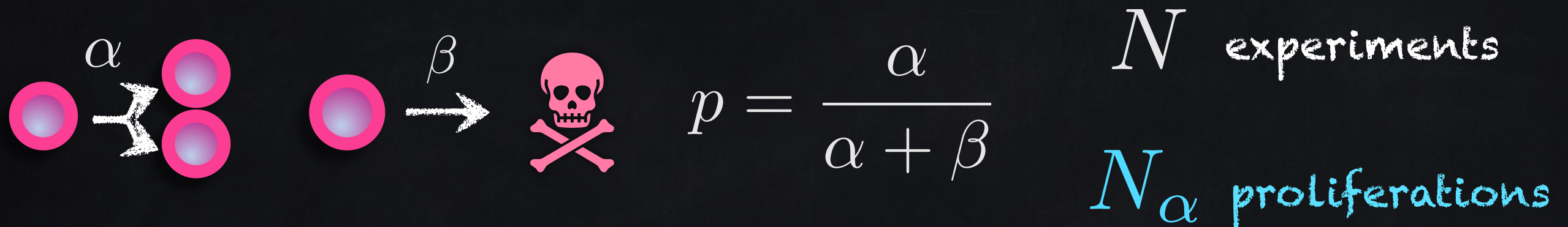
$$P(N_\alpha | p) = \binom{N}{N_\alpha} p^{N_\alpha} (1 - p)^{N - N_\alpha}$$

Likelihood



$$P(p | N_\alpha) = \frac{P(N_\alpha | p) P(p)}{P(N_\alpha)} \quad P(p) = 1$$

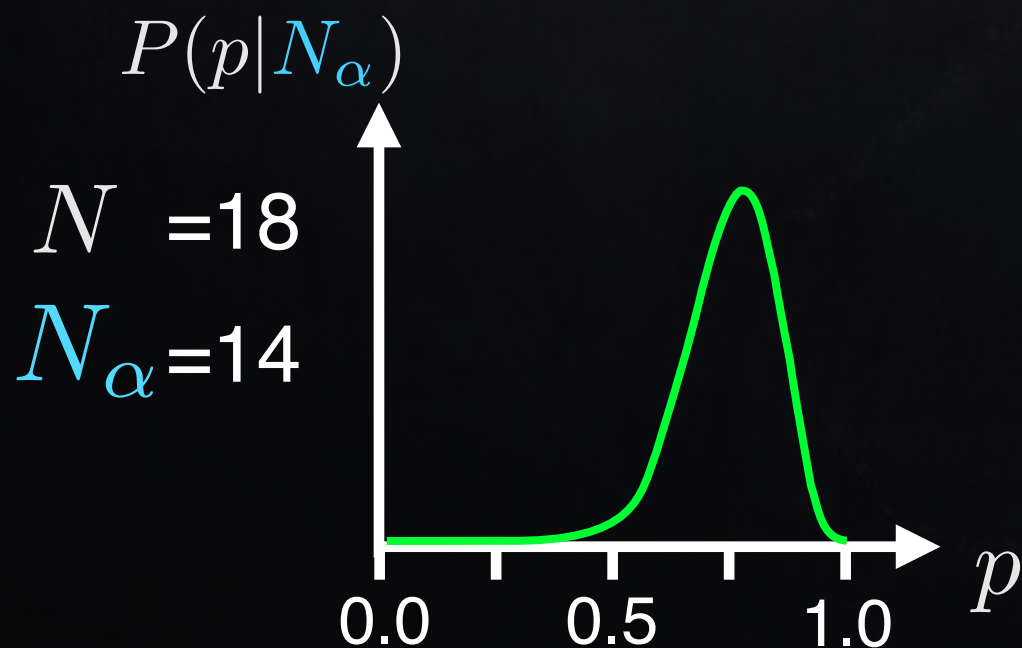
Example 1.1: Determining a proportion



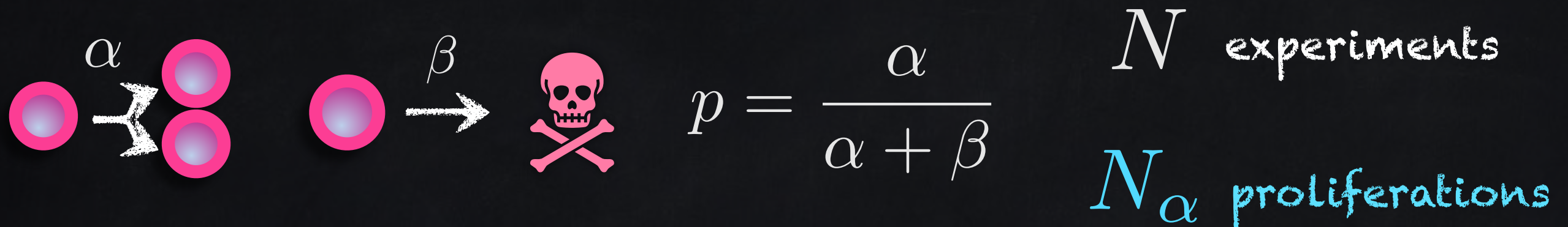
Posterior distribution

$$P(p|N_\alpha) = \frac{1}{P(N_\alpha)} \binom{N}{N_\alpha} p^{N_\alpha} (1-p)^{N-N_\alpha}$$

$$P(p|N_\alpha) \propto p^{N_\alpha} (1-p)^{N-N_\alpha}$$



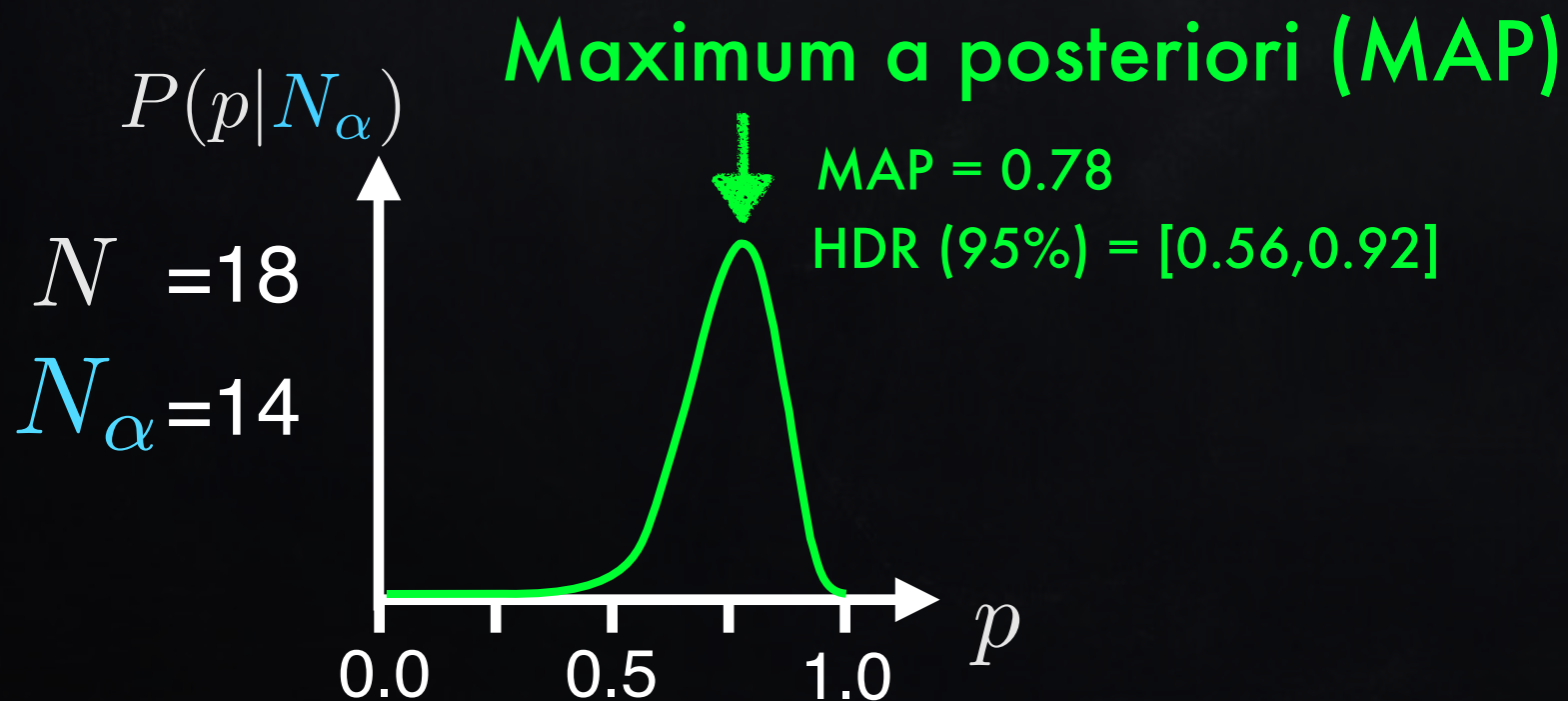
Example 1.1: Determining a proportion



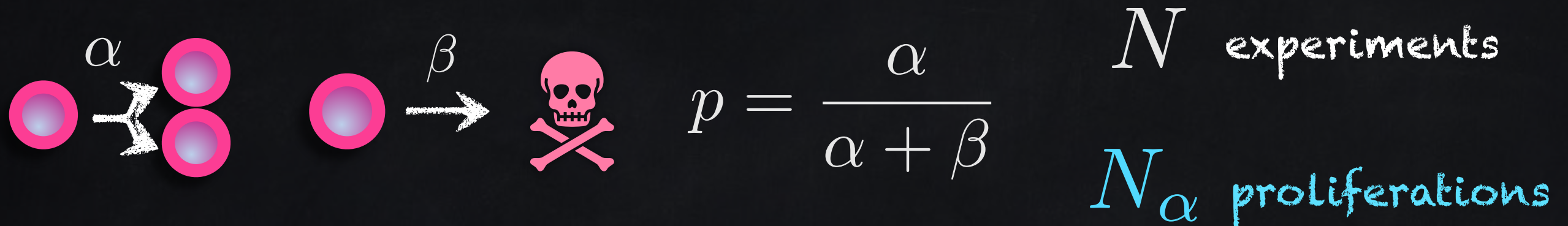
Posterior distribution

$$P(p|N_\alpha) = \frac{1}{P(N_\alpha)} \binom{N}{N_\alpha} p^{N_\alpha} (1-p)^{N-N_\alpha}$$

$$P(p|N_\alpha) \propto p^{N_\alpha} (1-p)^{N-N_\alpha}$$



Example 1.1: Determining a proportion

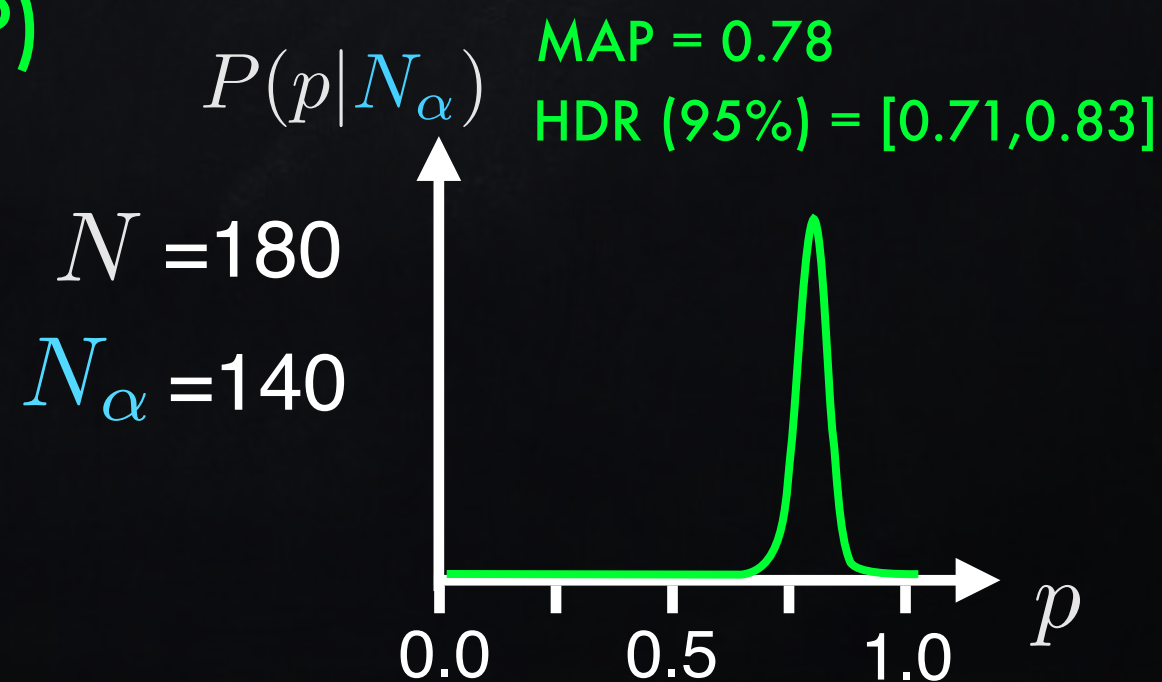
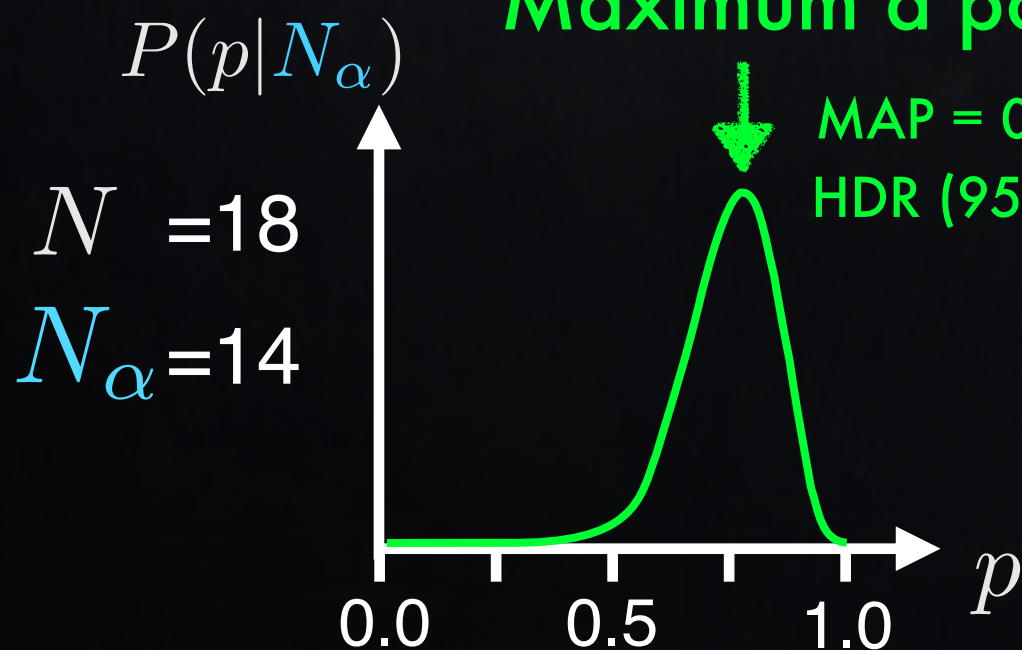


Posterior distribution

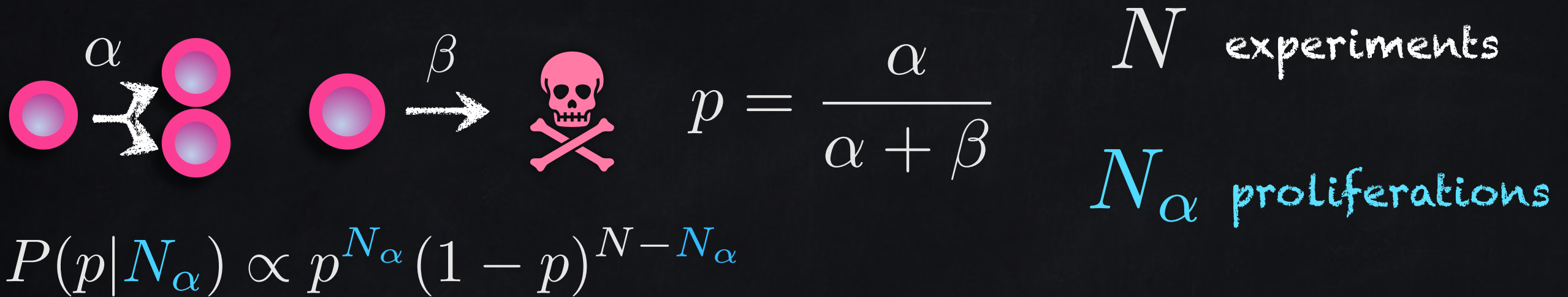
$$P(p|N_\alpha) = \frac{1}{P(N_\alpha)} \binom{N}{N_\alpha} p^{N_\alpha} (1-p)^{N-N_\alpha}$$

$$P(p|N_\alpha) \propto p^{N_\alpha} (1-p)^{N-N_\alpha}$$

Maximum a posteriori (MAP)

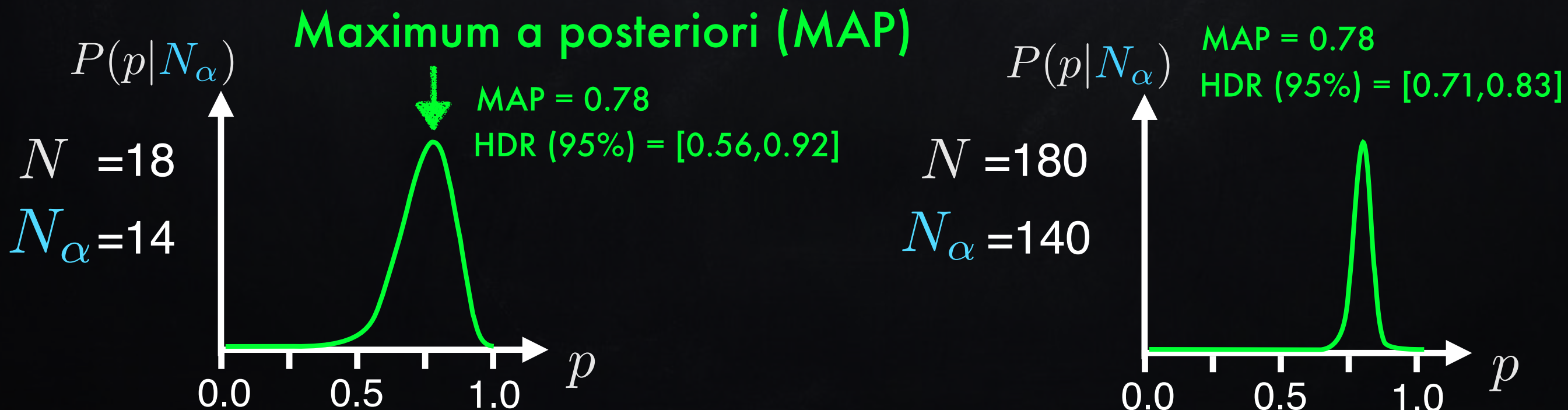


Example 1.1: Determining a proportion



Bayesian inference automatically incorporates the replicate information

Sometimes posteriors can be analysed analytically



Example 1.1: Determining a proportion

α β

$$p = \frac{\alpha}{\alpha + \beta}$$

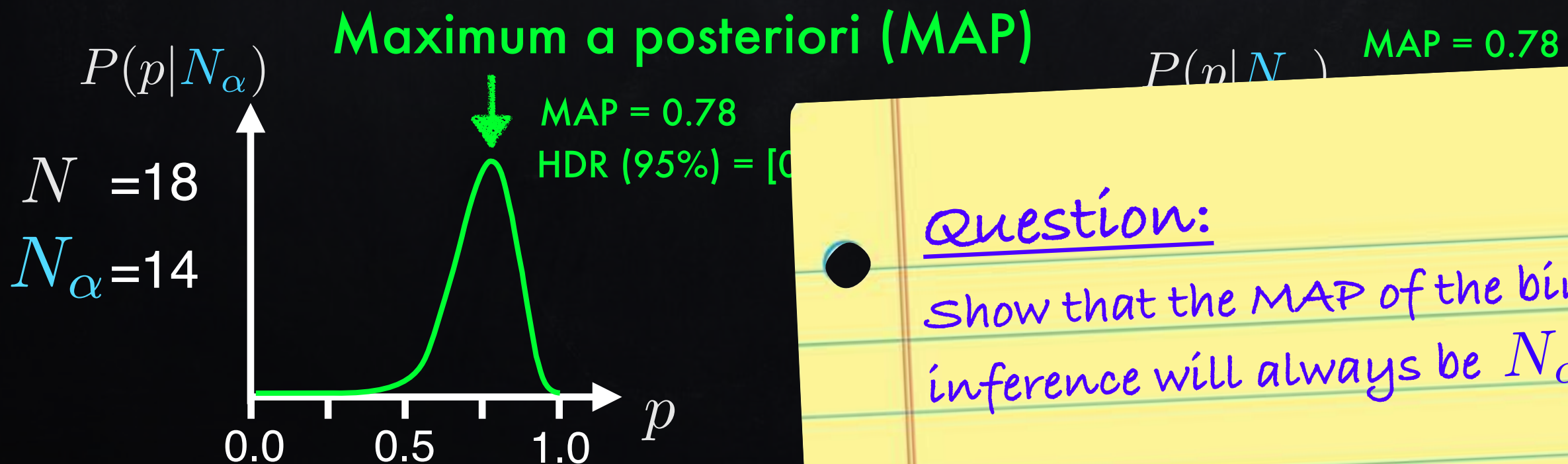
N experiments
 N_α proliferations

$$P(p|N_\alpha) \propto p^{N_\alpha} (1 - p)^{N - N_\alpha}$$



Bayesian inference automatically incorporates the replicate information

Sometimes posteriors can be analysed analytically



Question:

Show that the MAP of the binomial inference will always be N_α / N

Question:

Somebody before us has done the same experiment and has observed 10 proliferations in 30 experiments.

- How would you include this as a prior distribution?
- Will you obtain the same posterior distribution by including this previous experiment as a prior distribution than adding the information of both experiments in the likelihood and assuming a uniform prior?